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Section 1

Introduction

Managing personal money well has become much more important in today's economy. This is because of rising prices, high debt, and the increasing cost of living. Digital budgeting apps are now essential tools for tracking spending and planning how to use money for long term stability. Modern versions of these apps go beyond simple tracking. They use feedback and goal monitoring to help users make better financial choices (NerdWallet, 2026).

This study provides a detailed look at three Android budgeting apps which are YNAB, Wallet by BudgetBakers, and 22seven. These apps were picked because they use different methods and technical designs. YNAB uses a strict zero-based method to change behaviour while Wallet focuses on data and deep analytics. 22seven is different because it is made specifically for the South African banking system.

The main goal of this research is to check the features and flaws of each app to find the best parts for a new budgeting system. The study uses individual app reviews, a visual comparison table, and a list of recommended features. This method ensures that the final design for the proposed app is based on real research instead of just personal opinions. This creates a strong foundation for the technical and behavioural parts of the new system.

1.YNAB (You Need A Budget)

Overview

YNAB operates as a financial system built on zero based allocation. The platform follows four main rules which involve giving every Rand a job, planning for future costs, adjusting for changes, and reducing financial risk (YNAB LLC, 2024a). This method changes budgeting from just watching money to actively planning its use. Unlike apps that look at past spending, YNAB requires planning before any money is used (NerdWallet, 2026).

It includes normal features like categories and reports (YNAB LLC, 2024b). However, its real strength is how it limits spending. By making users plan before they buy, YNAB creates a strong habit of responsibility. This turns simple tracking into a strategic plan for the future.

Screenshots

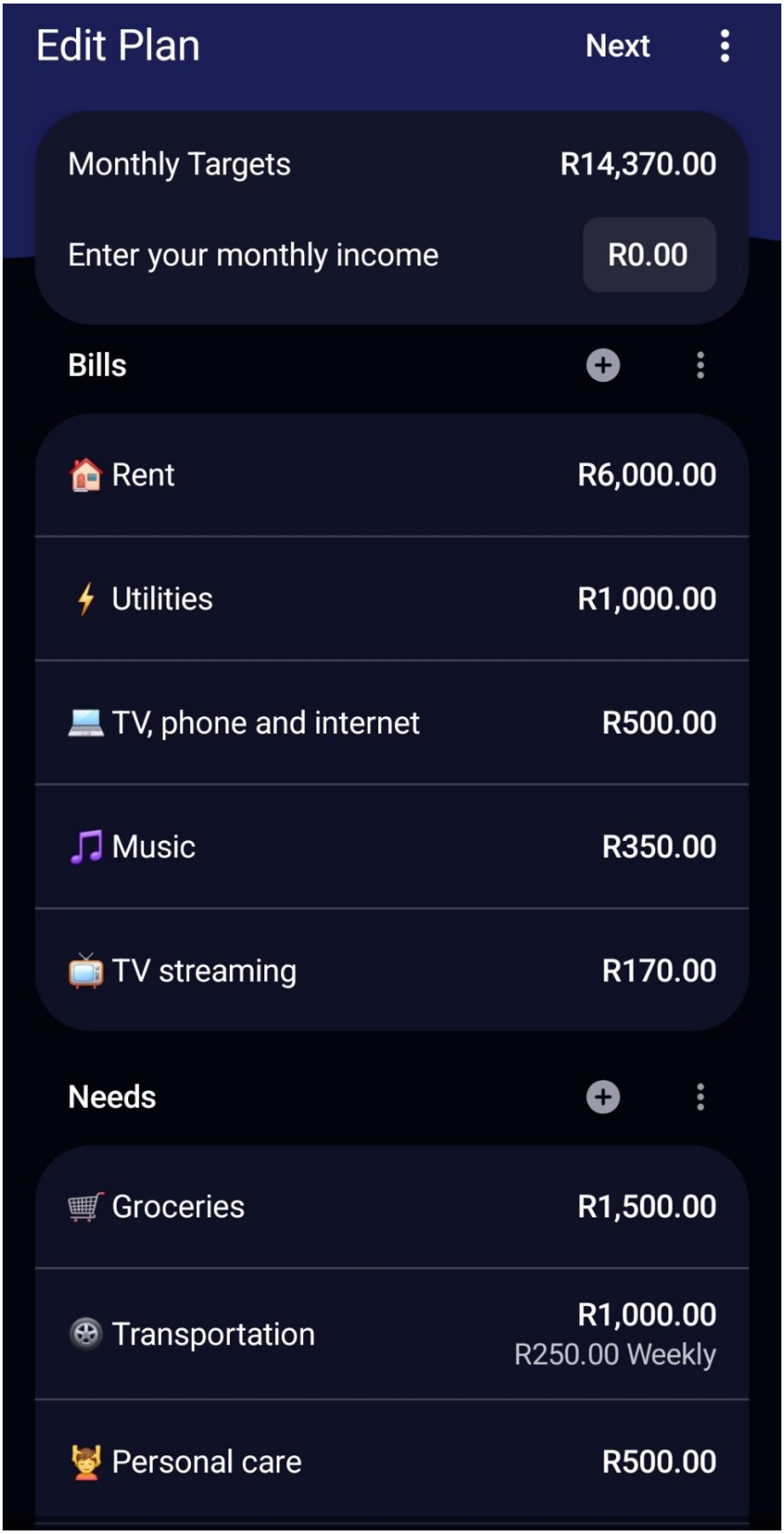

Figure 1: YNAB category-based monthly budgeting structure showing allocation across Bills and Needs categories.

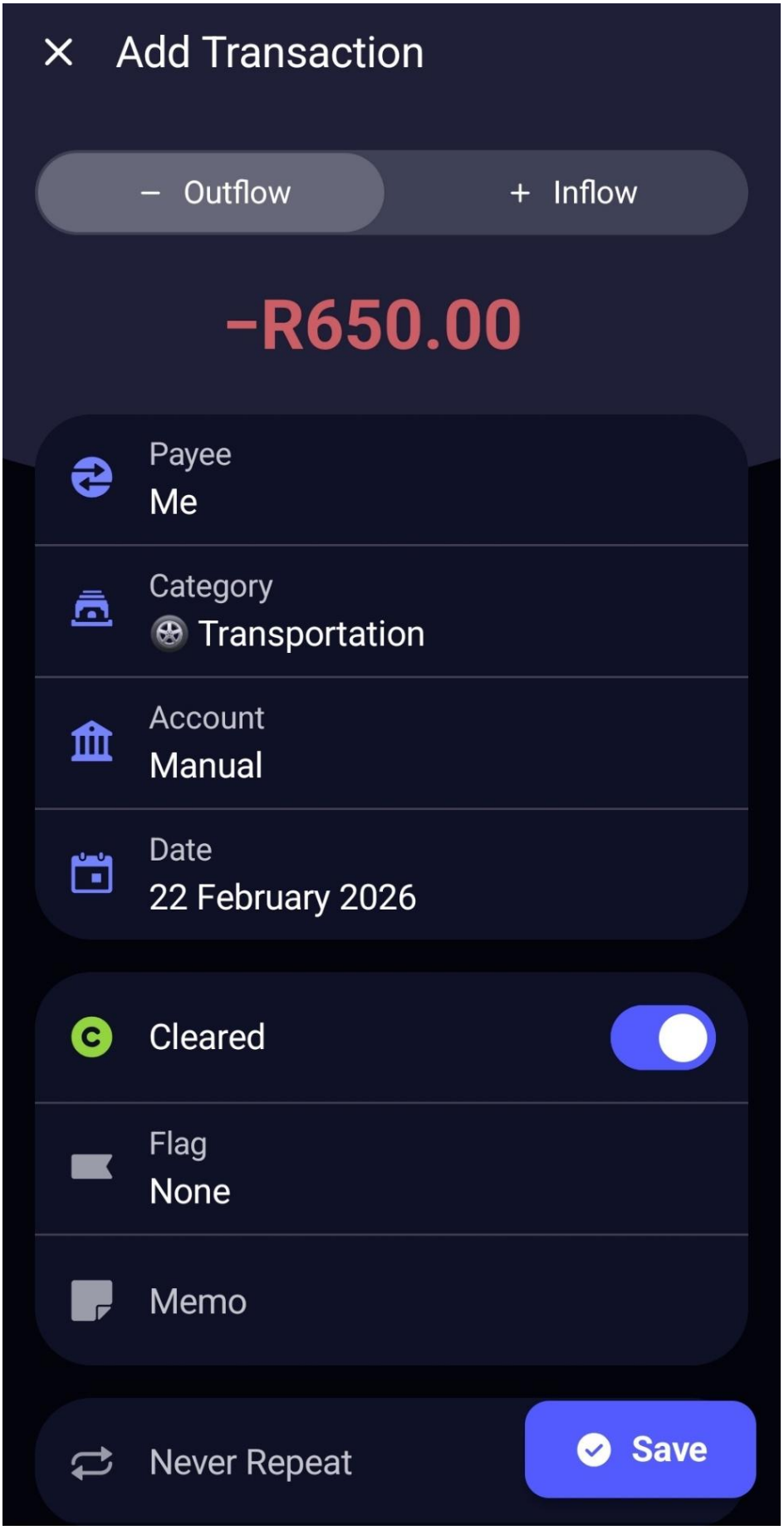

Figure 2: YNAB manual transaction entry interface demonstrating category assignment and real-time expense recording.

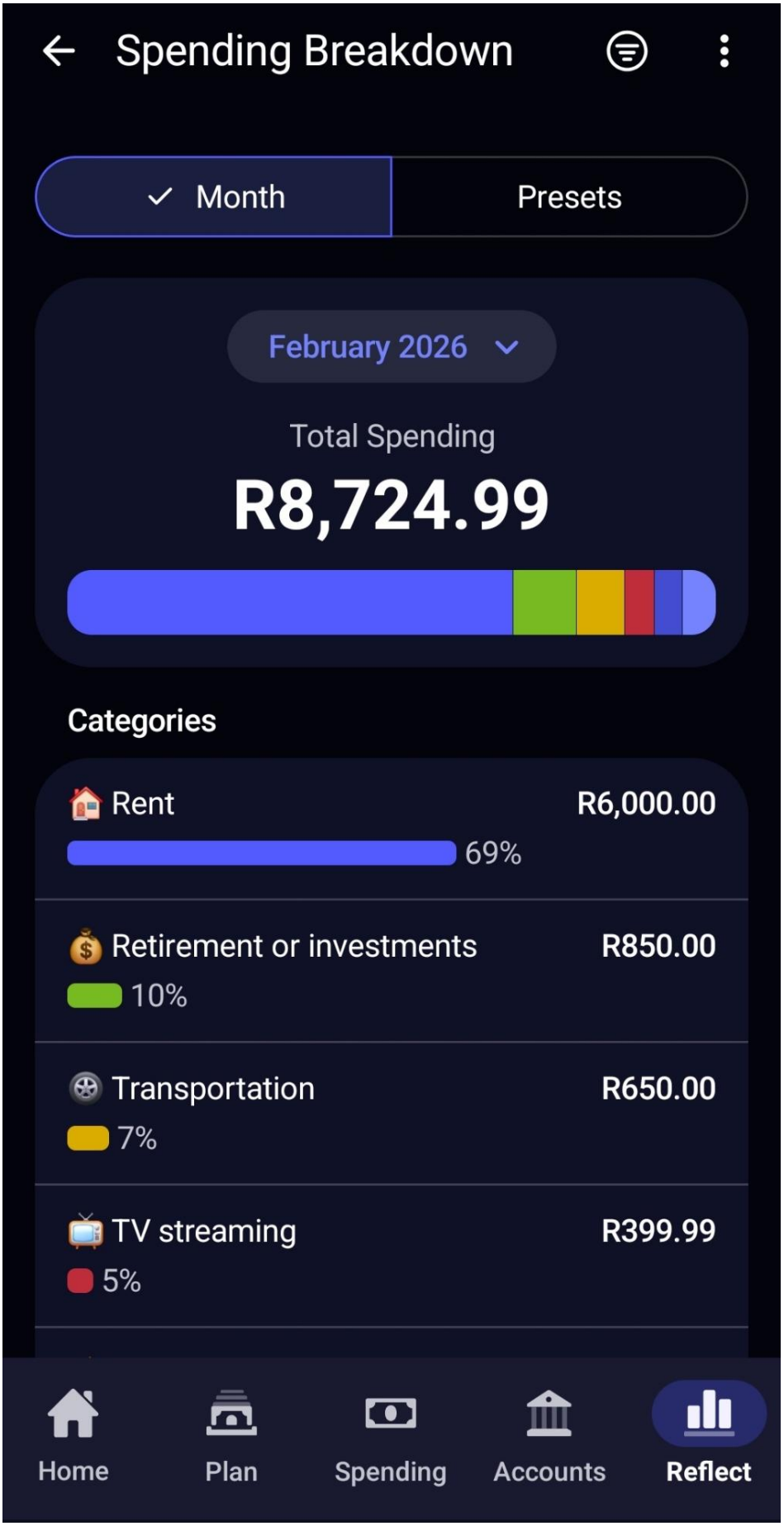

Figure 3: YNAB spending breakdown dashboard displaying category distribution and monthly expenditure trends.

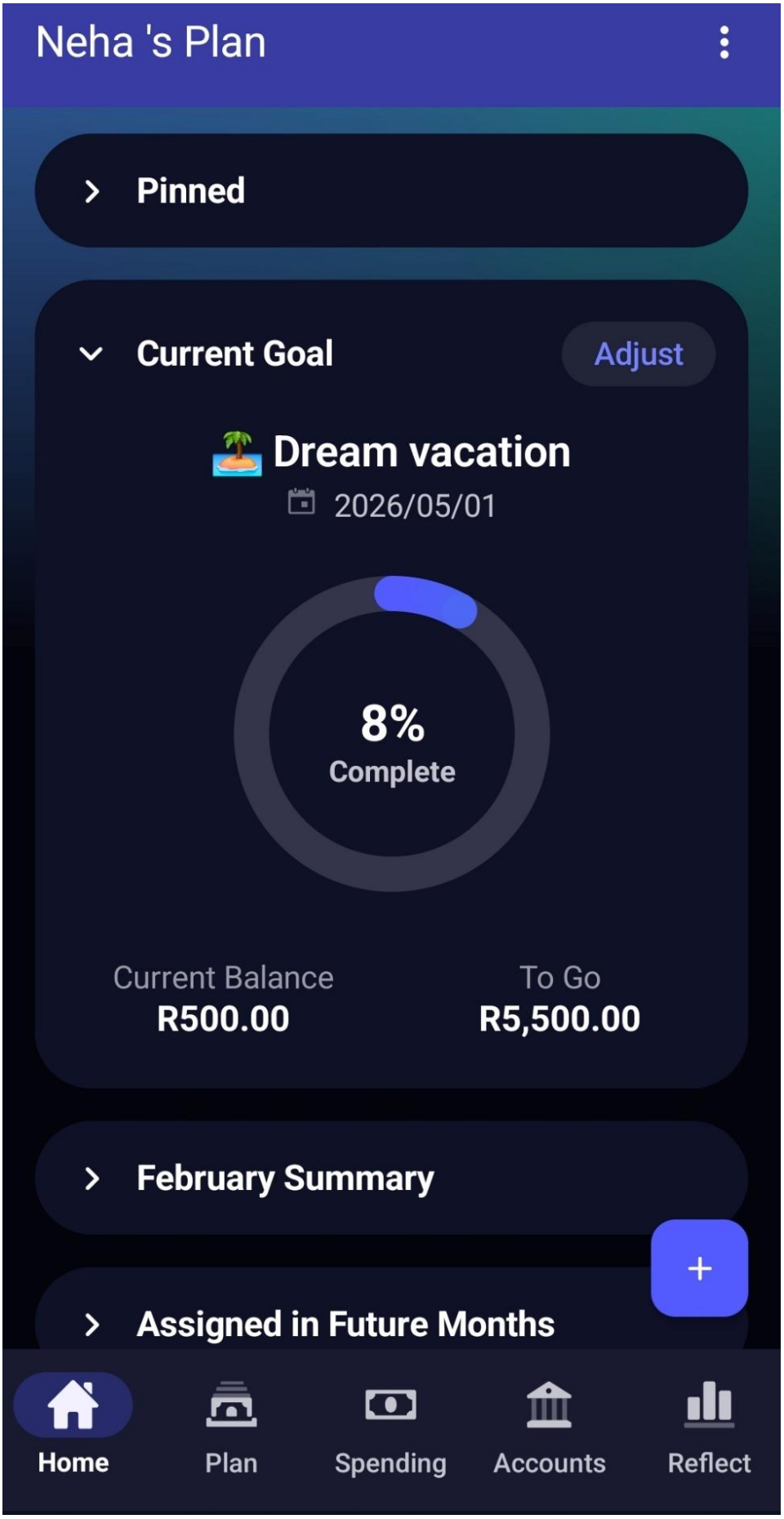

Figure 4: YNAB goal-tracking interface displaying savings progress and completion percentage.

Strengths

The main value of YNAB is its proactive design which requires users to name a purpose for their money before they spend it. This setup helps users think carefully about their choices and reduces impulsive spending (YNAB LLC, 2024a). Unlike other apps that only look back at spending, YNAB makes users responsible by making it part of the core budgeting method.

The tools in the app show category breakdowns and live balance updates to turn simple data into useful financial information (YNAB LLC, 2024b). Also, using goal tracking helps match small daily costs with long term plans to keep habits consistent. Together, these features make YNAB a system designed to change financial habits instead of just listing old transactions.

Weaknesses

Even though the app works well, YNAB has some usability issues that might stop more people from using it. Using a zero-based method requires constant mental effort and careful manual work to track every Rand. These factors often cause frustration, especially for users who are not used to strict financial systems (NerdWallet, 2026).

Also, the paid subscription plan creates a major barrier for people with lower incomes, such as students or those starting their first jobs. While the app allows users to move money between categories, its strict design can feel too controlling for people who prefer a more flexible approach. Because of this, YNAB chooses to focus on building self-discipline instead of providing automated ease or flexibility.

Innovative features

The biggest innovation in the YNAB system is how it reinforces good financial behaviour. The platform avoids just watching money and instead forces users to take action by naming a purpose for their capital before they spend it. This proactive model uses core ideas from behavioural finance right in the app design (YNAB LLC, 2024a).

The success of this method is improved by constant visual feedback. This is done through progress bars, category balance updates, and live dashboards (YNAB LLC, 2024b). These features create a loop that makes users constantly check and improve their financial choices. Unlike apps that focus on automation to make things easy, YNAB focuses on the ideas behind budgeting. Its main contribution is how it

changes the way users interact with their money by building discipline directly into the interface. This shift makes YNAB different from other apps and represents a major change in how financial tools are designed for people.

Relevance to our Proposed Application

The YNAB zero based allocation framework shows that naming a purpose for money early is very effective. This promotes proactive planning and helps stop reactive spending. The use of live balance updates and progress bars shows how visual feedback can improve user awareness. Adding similar category monitoring and goal tracking tools to our proposed app will help make users more accountable.

However, the difficult learning curve of the YNAB model means we must design an app that balances strict rules with easy setup steps. This will make the system more accessible for new users while keeping the focus on disciplined financial habits.

2.Wallet by BudgetBakers

Overview

Wallet, created by BudgetBakers s.r.o., works as a financial platform that helps with complete monitoring through many different features. Its design includes category based budgeting, detailed transaction logs, digital receipt storage, the scheduling of regular payments, and complex financial reporting (BudgetBakers, 2024).

Unlike the strict and proactive rules used by YNAB, Wallet uses a design that focuses on visibility. This method values clear analysis and user awareness over strict rules for behaviour. The main interface combines different account balances and spending trends to help users see complex patterns while staying flexible. While this open structure gives users more freedom, it does not enforce discipline as strongly as zero based systems. Because of this, Wallet shows a major challenge in financial technology design which is the balance between being flexible and being disciplined. This balance is a key part of how our proposed app will be built.

Screenshots

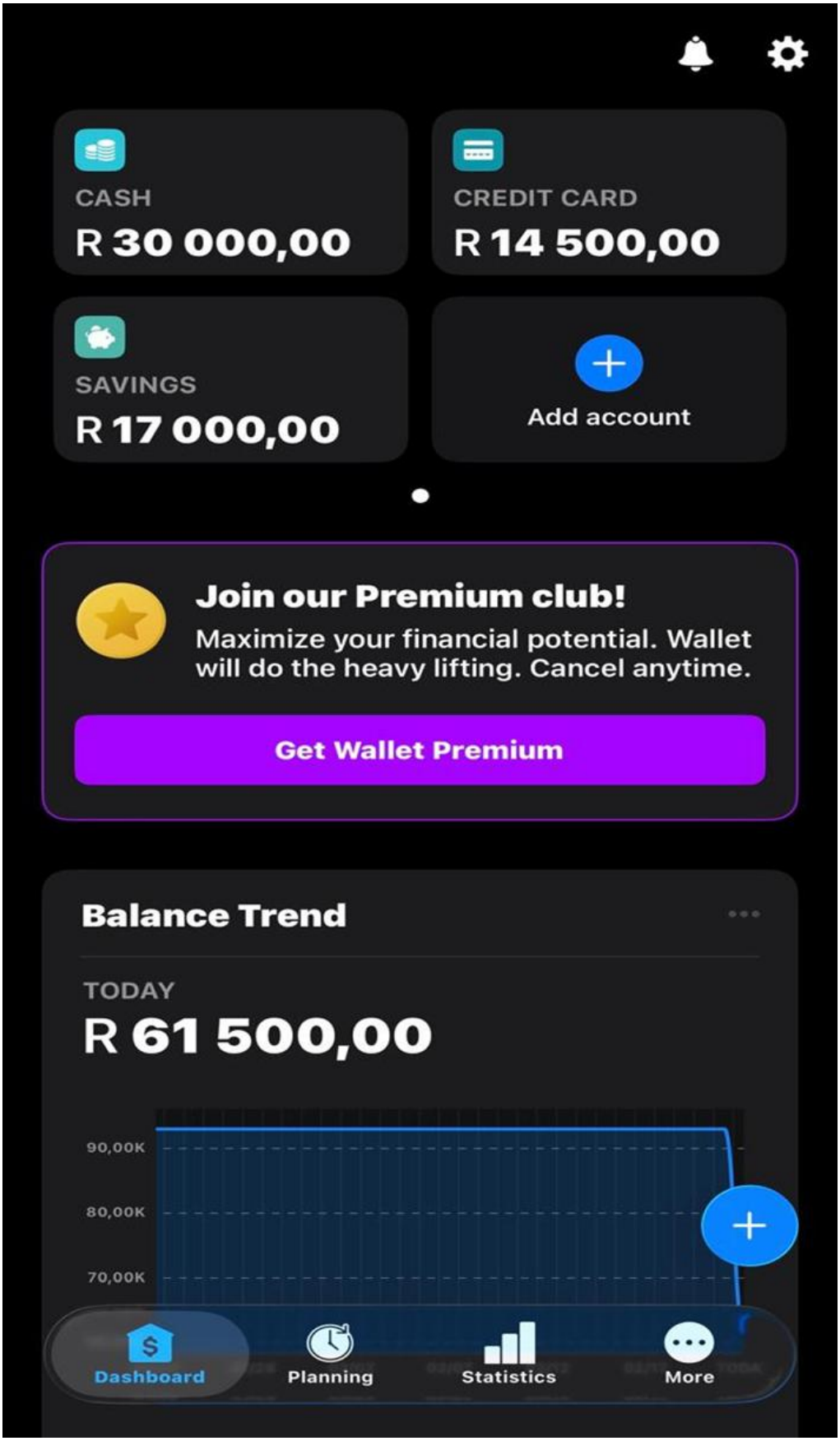

Figure 5: Wallet Dashboard illustrating financial aggregation, embedded premium upsell integration, and historical balance trend visual analytics.

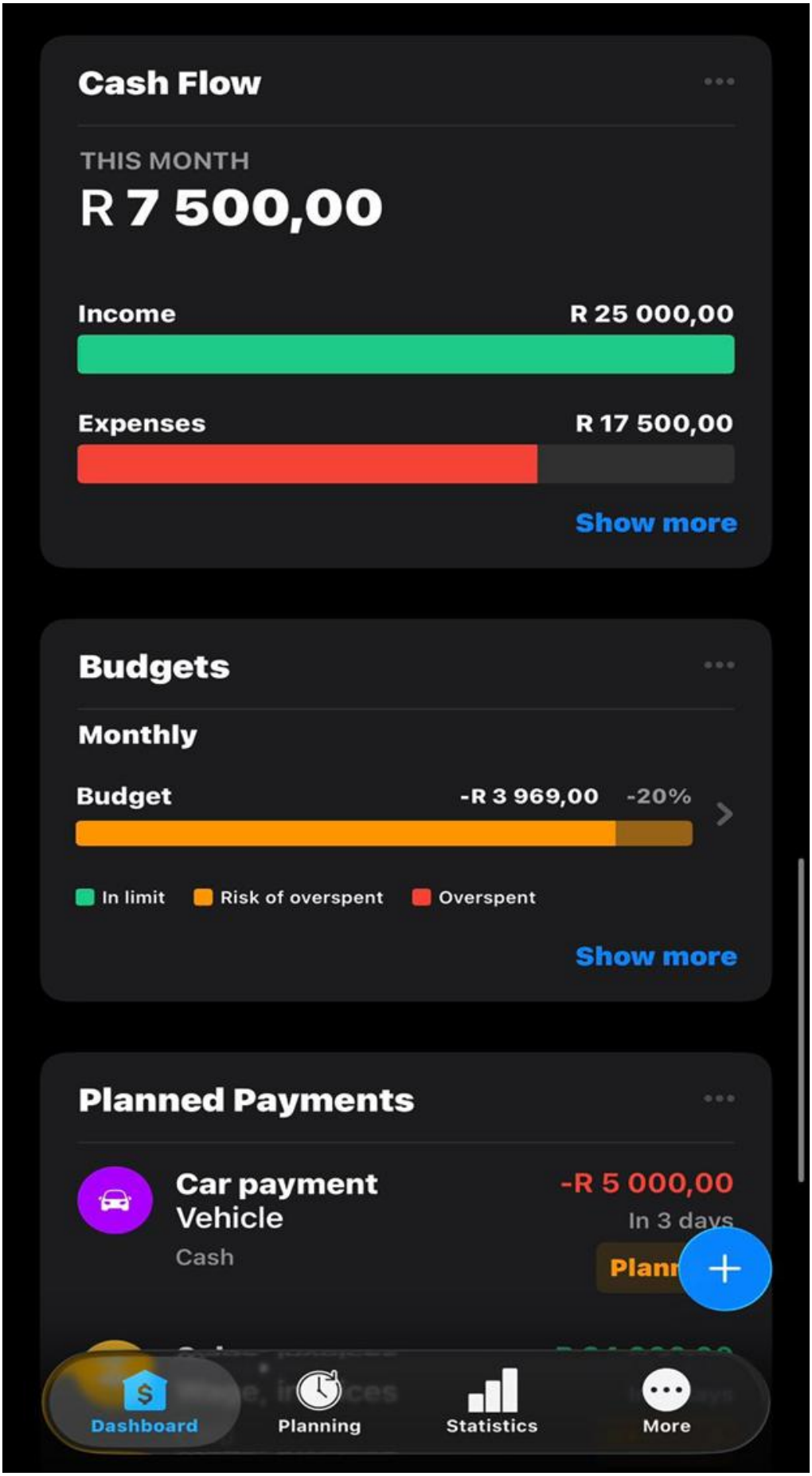

Figure 6: Wallet interface demonstrating automated cash flow calculation, overspending risk indicators and proactive financial planning features.

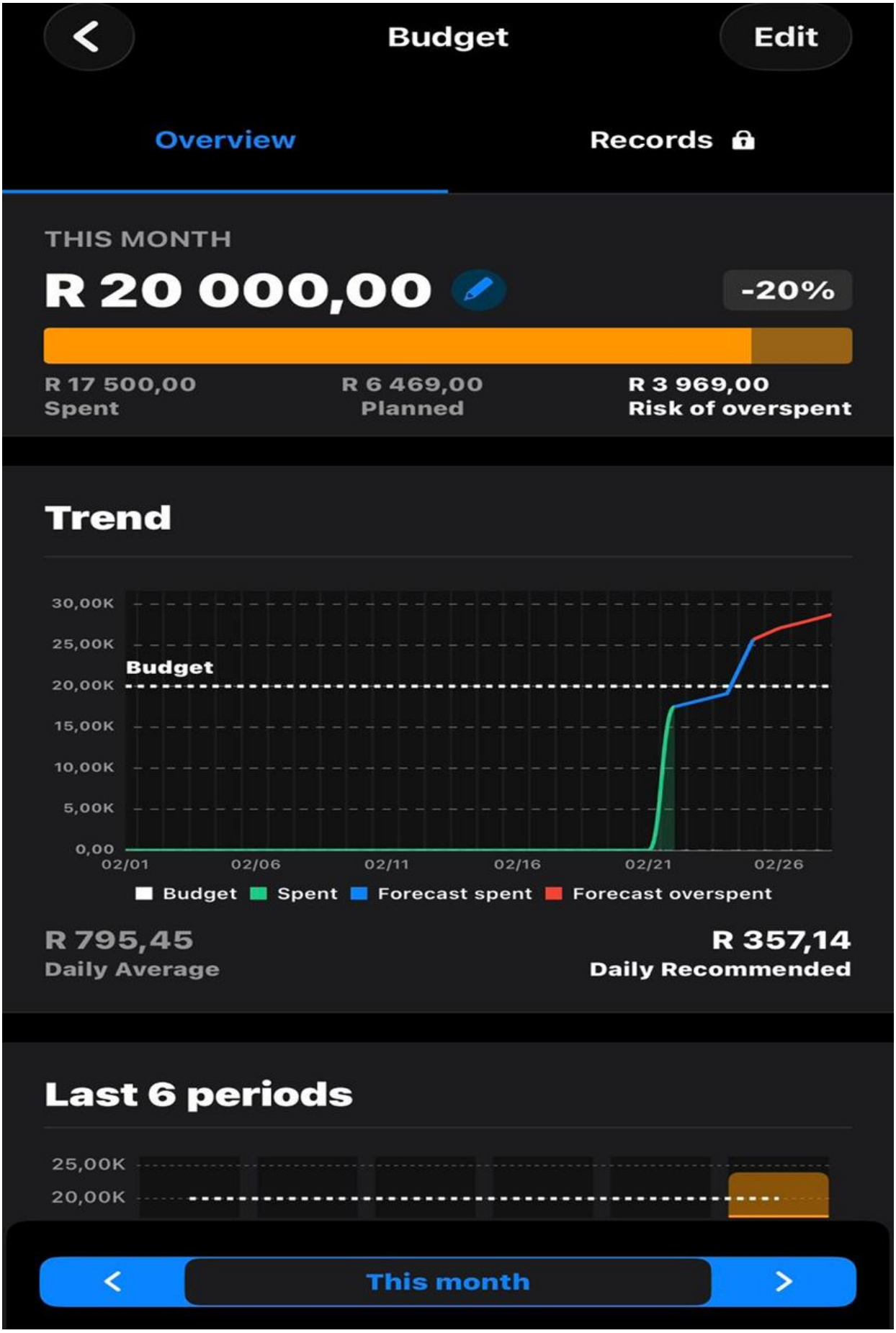

Figure 7: Wallet Budget overview presenting predictive forecasting functionality and real-time expenditure tracking mechanisms.

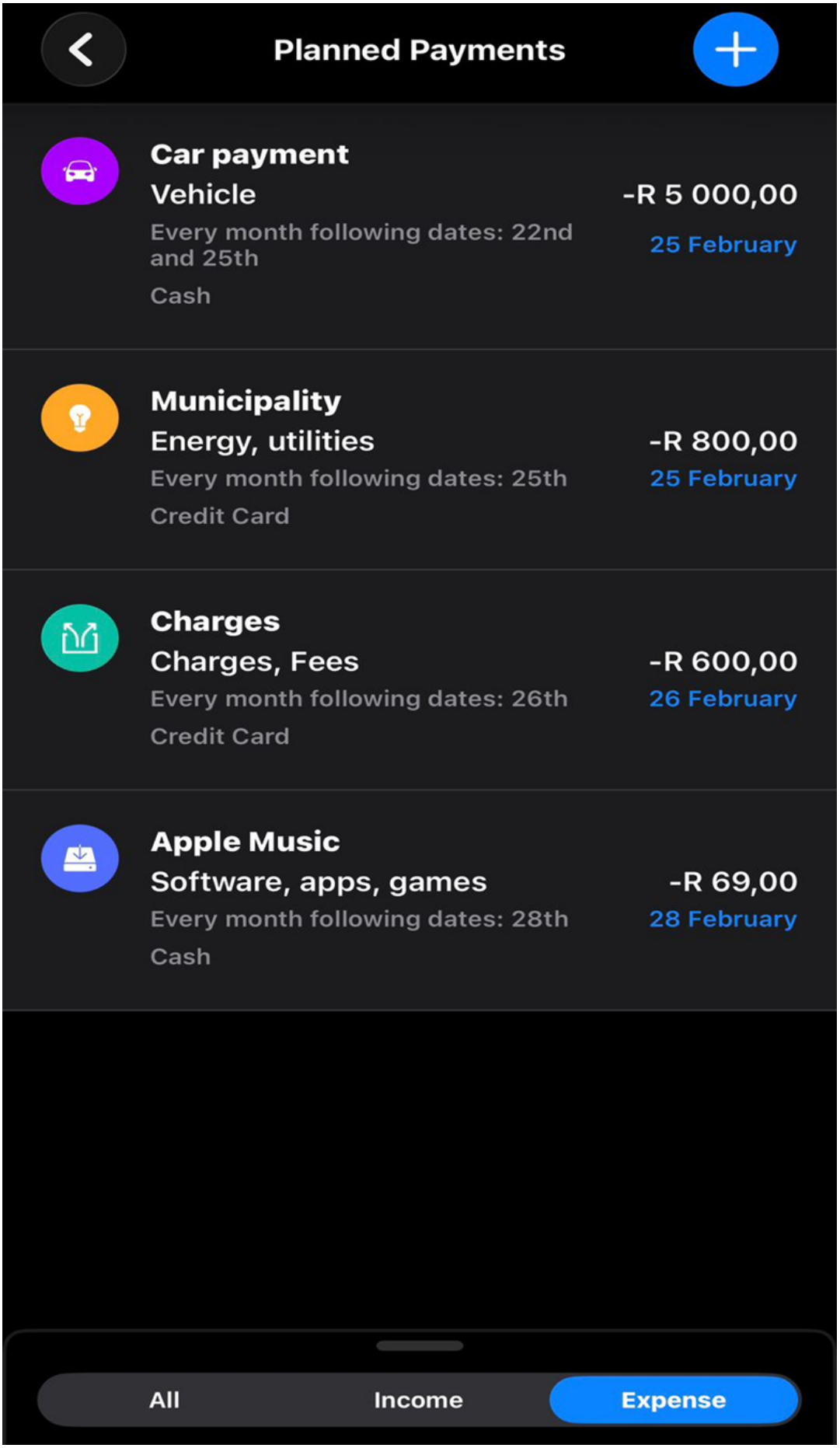

Figure 8: Wallet Planned Payments module highlighting recurring transaction automation and structured financial liability management.

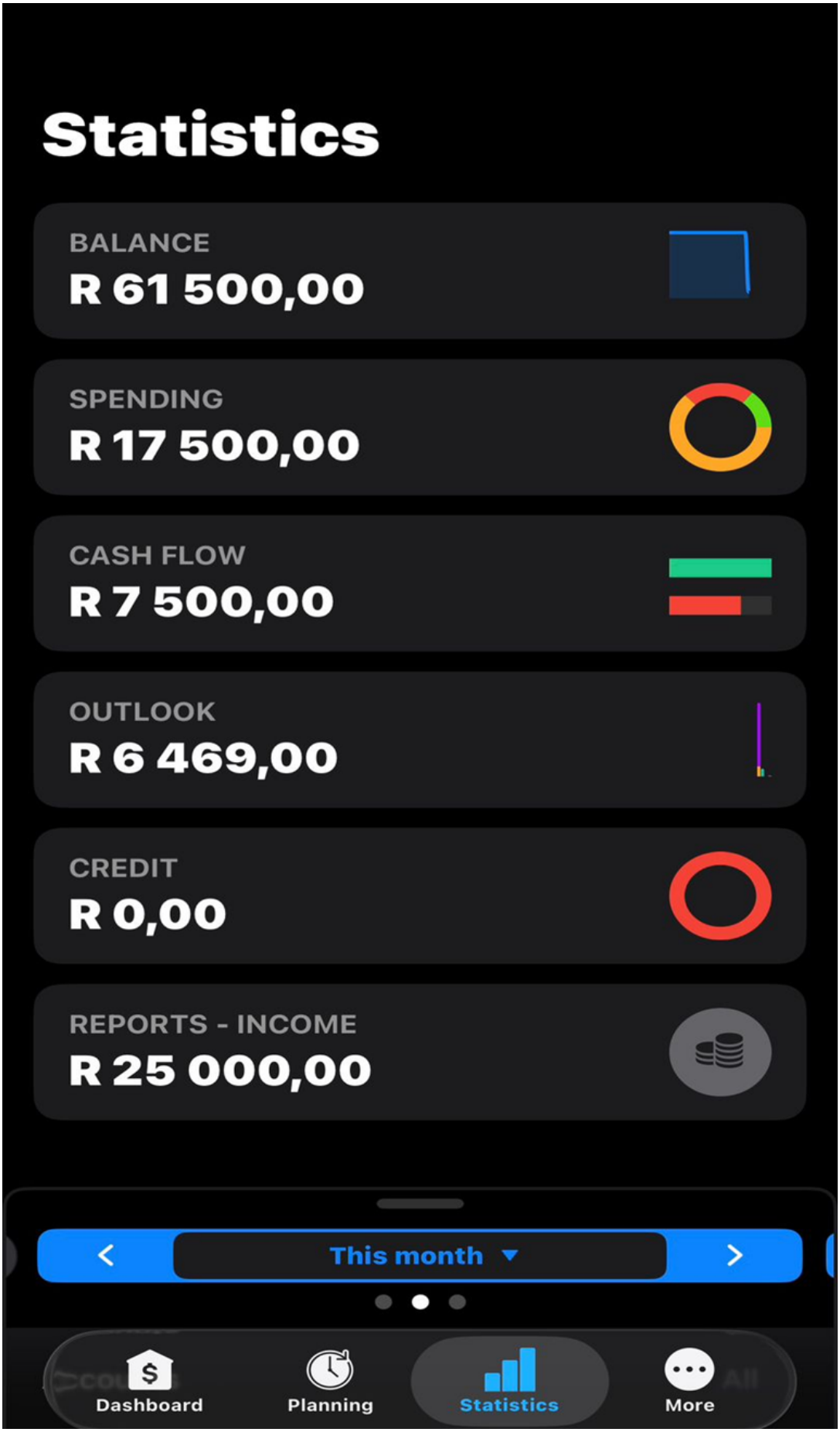

Figure 9: Wallet Statistics dashboard consolidating multi-dimensional financial performance indicators.

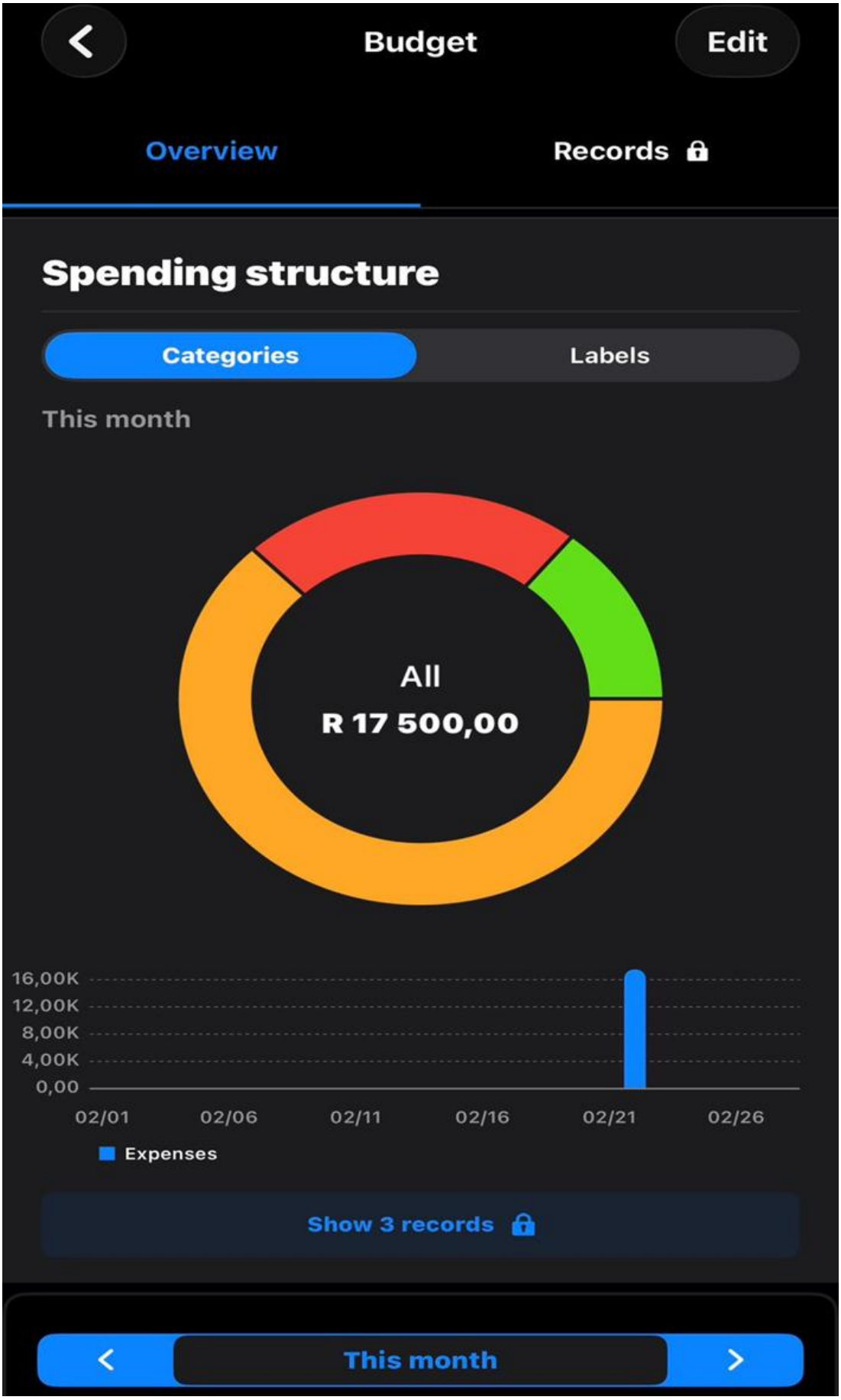

Figure 10: Wallet Spending Structure screen visualising categorical expenditure analysis and trend identification support.

Strengths

The main value of the Wallet system is its clear analysis and consistent design. By using different time filters and advanced graphs, the app helps users see spending patterns and make better financial changes (Sahni et al., 2023).

The detailed sorting of transactions and the ability to upload digital receipts helps users stay responsible and think more about their money (Kempson and Poppe, 2021). Also, setting up regular payments for bills creates a plan for future costs which helps build financial strength (OECD, 2022). The app also allows for automatic bank updates and shared accounts which shows it can grow and work with other systems (PwC, 2023). Together, these features help users become more aware of their money by keeping them involved in every part of their financial life.

Weaknesses

Despite its analytical benefits, Wallet has several structural and socio-economic limits. Restricting advanced features, like automated bank updates, to a paid subscription creates a barrier for people with low incomes (World Bank, 2021). Also, the app relies on third party connections for data which can cause worries about privacy and trust (OECD, 2022).

From a mental perspective, the focus on manual data entry can be hard for new users. This might make the app difficult to use and lead to people stopping (OECD, 2022). Furthermore, the lack of rewards or game like incentives can make it harder for users to stay motivated over time (Ranyard et al., 2022). Because of this, Wallet chooses to focus on being flexible and clear instead of being easy to use or building deep habits.

Innovative features

The main innovation of the Wallet platform is its integrated financial design. By combining structured budgeting, the scheduling of regular payments, and long-term data analysis into one system, the platform effectively centralizes money management and reduces the problems of scattered data (PwC, 2023).

Unlike platforms that focus entirely on automation, the innovation of Wallet is found in its architecture and design. This approach creates a consistent system that helps users think more about their money and encourages intentional engagement with their personal capital. Instead of making the user a passive observer of automated tasks, the design of the platform requires active participation. This involvement helps improve financial literacy and long-term responsibility.

Relevance to our Proposed Application

Wallet acts as a strong model by showing how live dashboards and long-term reports help users think more about their money. This design successfully moves users from just watching their spending to actively managing it.

Even with these strengths, our proposed system will improve on the limits of Wallet by being more inclusive. We will do this by removing the cost for core features so that more people can use them. By making the interface easier to use and adding reward systems, our design will focus on keeping users involved while maintaining a disciplined financial structure.

3.22seven (Vault22)

Overview

22seven is a well-known South African financial system that uses automatic bank updates, smart sorting of transactions, and combined dashboard reports (Vault22, 2024). Unlike the strict rules used by YNAB or the reflective design of Wallet, 22seven focuses on automation. Within this model, money transactions are pulled in and sorted automatically which reduces the need for manual data entry.

While this design makes the app very easy to use and accessible, it might also reduce how much users think about their spending compared to stricter systems. Because of this, the platform values making things easy over changing the core financial habits of the user.

Screenshots

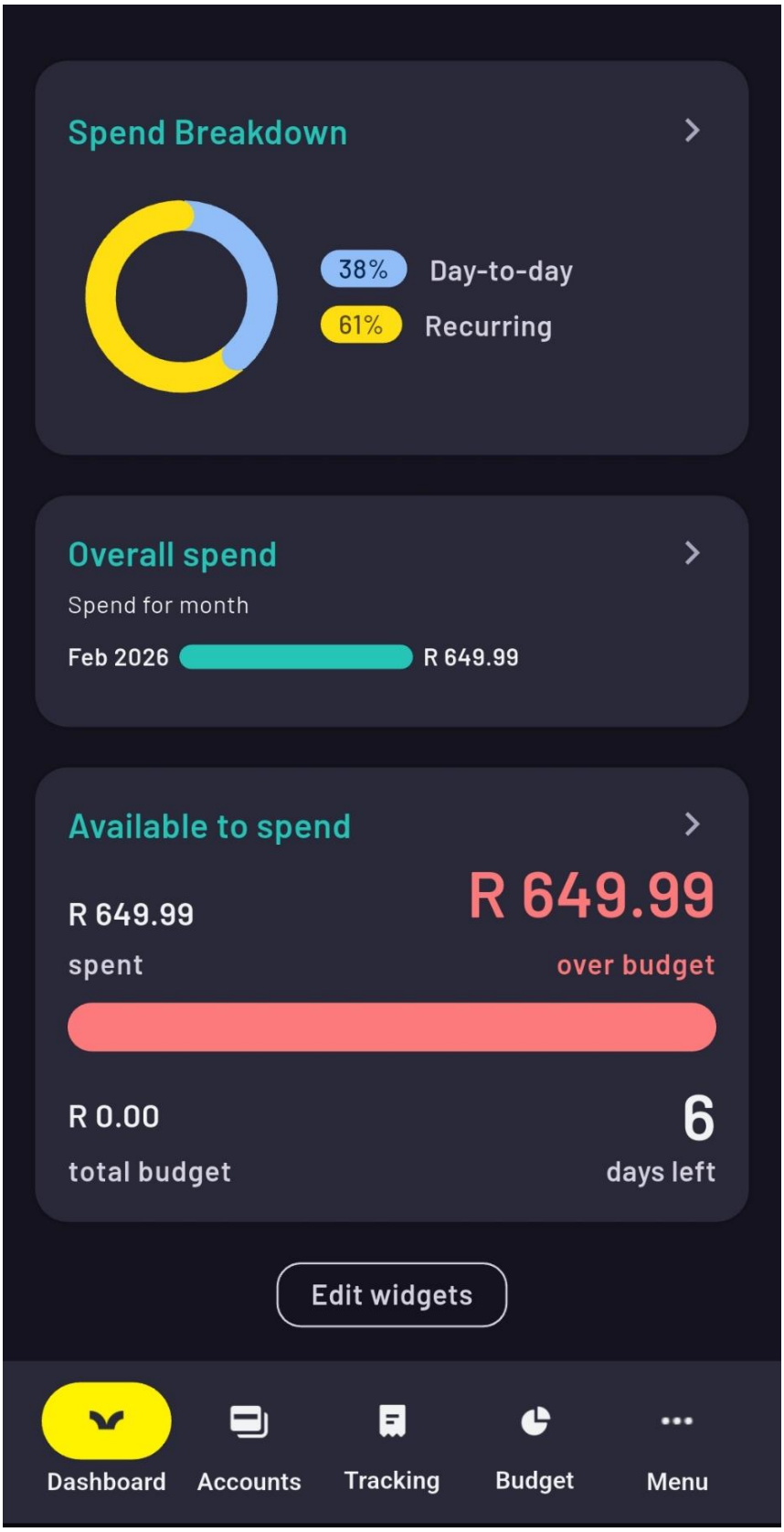

Figure 11: Vault22 Dashboard illustrating consolidated spend breakdown, monthly expenditure tracking and behavioural overspending alerts.

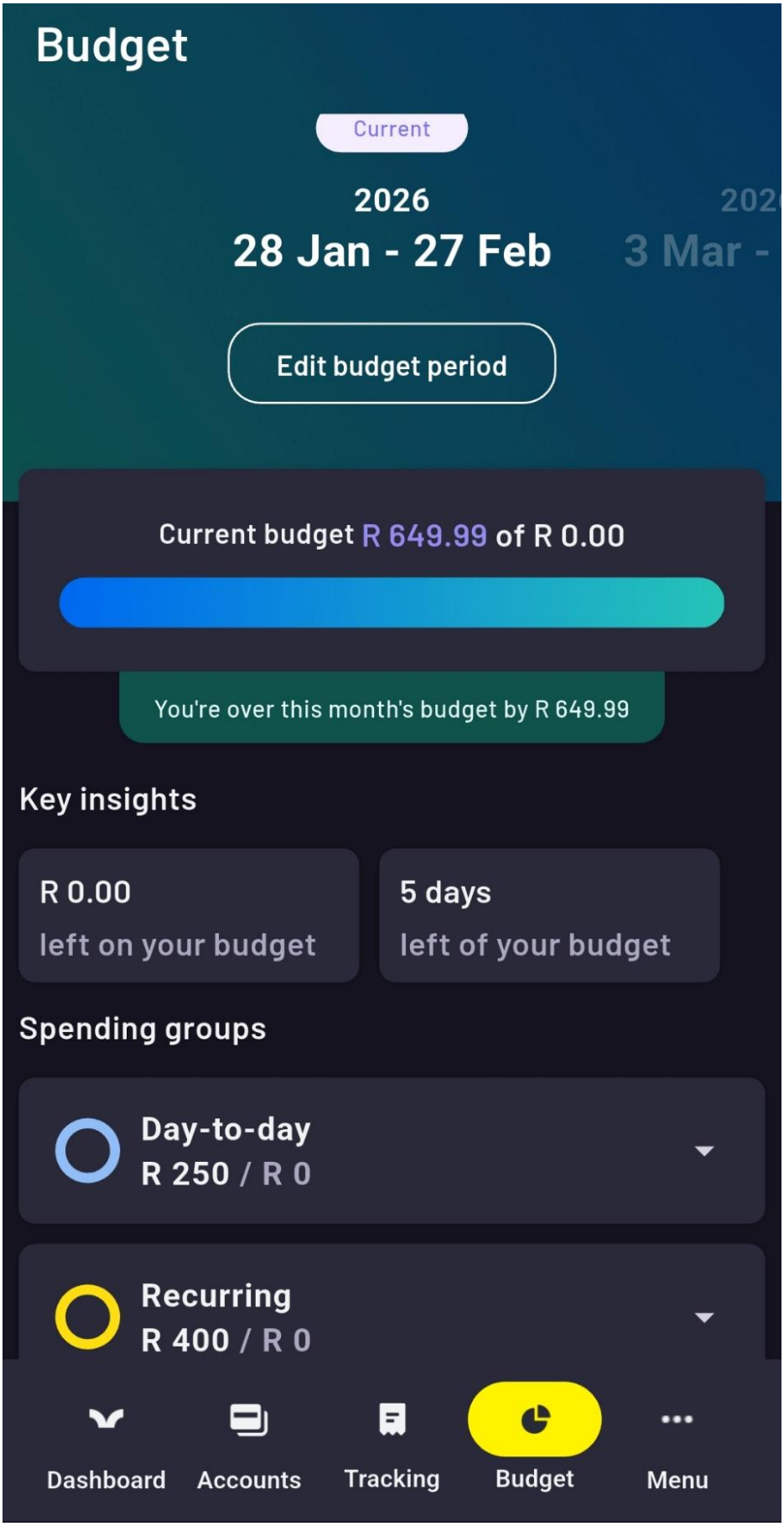

Figure 12: Vault22 Budget screen demonstrating automated budget monitoring and reactive feedback through visual progress indicators.

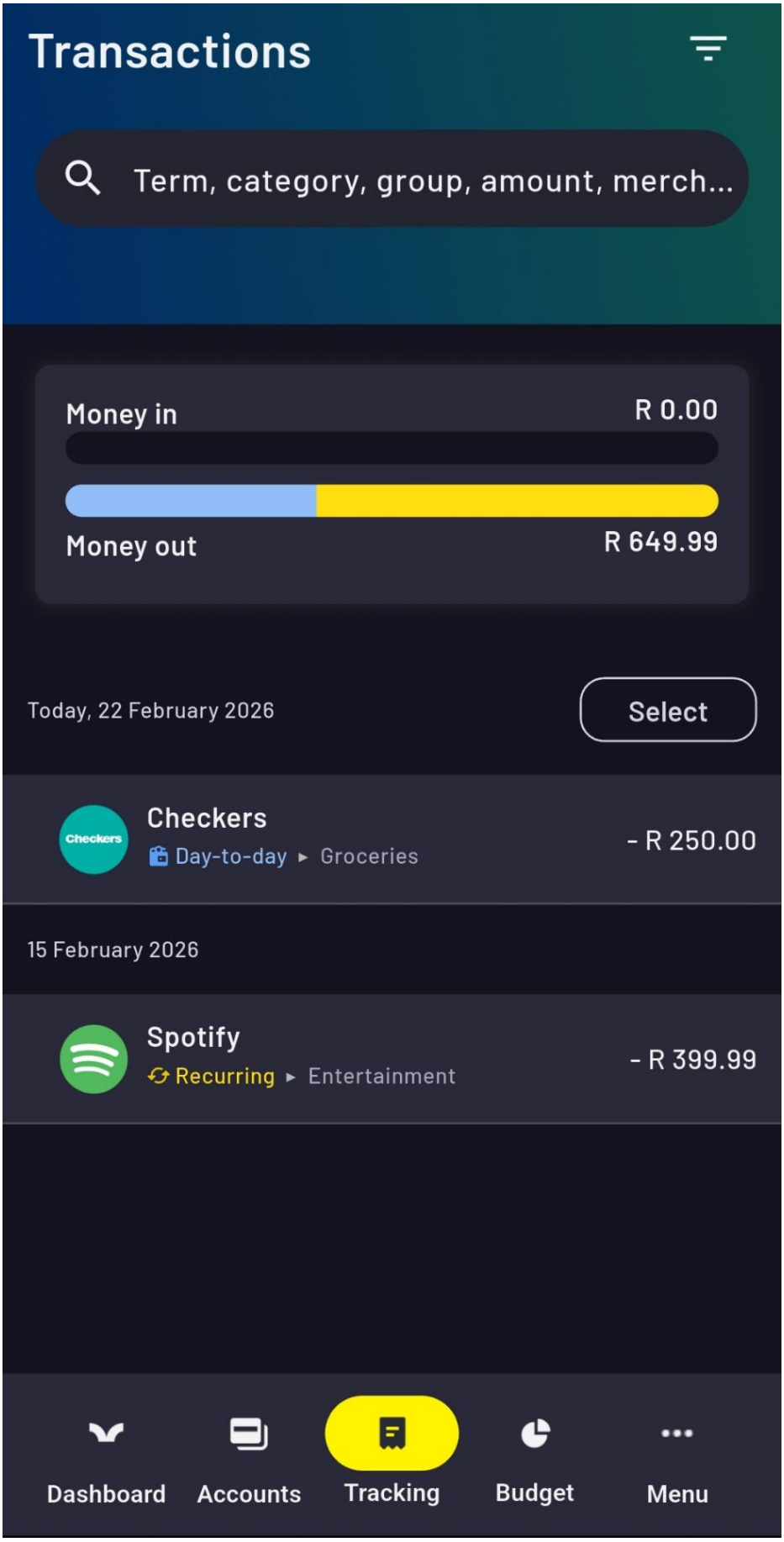

Figure 13: Vault22 Transactions interface highlighting automated categorisation and friction-reducing financial aggregation.

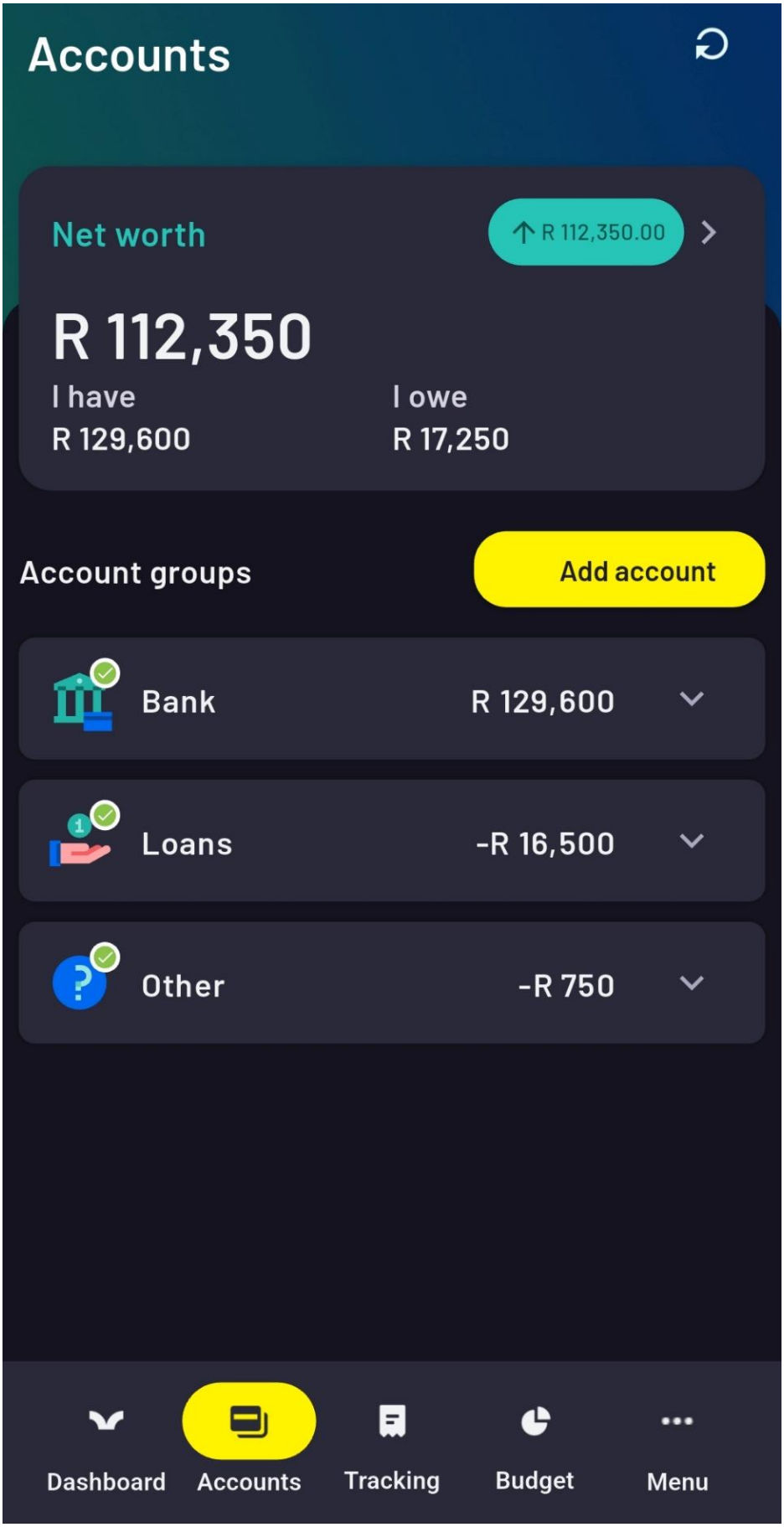

Figure 14: Vault22 Accounts overview presenting net worth calculation through multi-account financial aggregation.

Strengths

The main value of the 22seven platform is its ability to automatically group financial data. This feature combines different bank accounts into one clear dashboard so users can see all their money in one place (PwC, 2023). This single interface gives a complete view of a person's finances and helps reduce the confusion often caused by using multiple banks.

The platform also uses a behavioural engine to send digital nudges. These are timely alerts designed to help users fix their spending habits when needed (Ranyard et al., 2022). A major advantage of the app is that it works directly with South African banks. This makes the tool more relevant to local users and helps build trust within the regional market. Together, these design choices make 22seven a very effective and easy to use tool for broad financial monitoring.

Weaknesses

The platform depends on constant internet and stable connections with other banks which creates a technical risk. Any service break or delay directly hurts how well the app works and how complete the data is. This means the reliability of the system is tied to outside infrastructure instead of just the app itself.

Also, using automatic sorting might reduce how much control users have over their financial data. This lack of manual work can lower how much users think about their money. Research in behavioural finance shows that active participation is needed to truly understand financial data (Fernandes et al., 2021). By automating everything, the app might accidentally make users feel less responsible for their finances. Like Wallet, 22seven does not use rewards or game like systems to keep users motivated. Without these incentives, it is harder for users to keep up good habits over time. Because of this, 22seven chooses to focus on being fast and easy instead of building deep financial habits.

Innovative features

The main innovation of the 22seven platform is its advanced automatic financial grouping and its behavioural alert system. By combining data from many different banks into one secure interface, the platform greatly improves financial visibility within the South African market (PwC, 2023). This centralized system fixes the

problem of scattered information that often happens when people use multiple bank accounts.

Furthermore, the use of automatic transaction sorting and live spending alerts works to put the ideas of behavioural nudging into practice. This is done through a design that makes the app easy to use and tries to influence user choices with timely notifications instead of strict rules. Because of this, 22seven represents a new way of helping users manage their money by using technology to keep them aware of their finances without needing a lot of manual work.

Relevance to our Proposed Application

The 22seven ecosystem shows how automatic sorting and live behaviour alerts can make a platform easier to use for beginners. These features help lower the barriers for people who are new to budgeting apps. However, our proposed application is designed to move between automatic efficiency and active user involvement to prevent users from becoming detached from their finances.

By combining the structured self-regulation of Wallet with the operational speed of 22seven, our system aims to balance deep behavioural habits with easy access. This method uses the best parts of both design styles to create a strong foundation for an inclusive financial management system.

Visual Comparison

The visual infographic provides a clear and organized way to compare the different budgeting methods and technical designs of the three apps. This summary makes it easy to see where the platforms are similar and where they differ. By looking at these differences, we can create a strong and research-based reason for the design choices in our new system.

By putting all these details into a single comparison table, the study shows the technical trade-offs found in each platform. This organized approach ensures that our final design is not just based on what we like but is the result of checking the best practices in modern financial technology. Because of this, the infographic acts as a guide for building a more inclusive and effective financial management tool.

Comparison of Budgeting Applications: YNAB, Wallet and 22seven

YNAB (Behavioural Budgeting)

Strengths

- Zero-based budgeting method
- Strong category limits and planning
- Budget progress tracking

Comparison

- More structured than Wallet and 22seven
- Similar expense tracking and reporting features

Wallet (Analytics & Visualisation)

Strengths

- Strong dashboards and graphs
- Automatic expense categorisation
- Overspending alerts

Comparison

- Easier to use than YNAB
- Similar budgeting categories and tracking features

22seven (SA Financial Insight)

Strengths

- South African bank integration
- Automated spending insights
- Very beginner-friendly

Comparison

- Stronger regional insight than the other apps
- Shares alerts and reporting capabilities with Wallet and YNAB

Shared features: All three applications support expense tracking, category budgeting and financial reporting. YNAB provides budgeting discipline, Wallet excels in visual analytics, and 22seven offers regional financial insight. These combined strengths inform the design of the proposed application.

Figure 15: Comparative overview of selected budgeting applications highlighting similarities and differences in budgeting methodology, usability and financial insight features.

The visual cross analysis shows that while all three apps share basic features like category monitoring, dashboards, and financial reports, they are very different in how they manage user behaviour and automation. YNAB uses a system of planning ahead and strict discipline while Wallet focuses on clear analysis and user oversight. In contrast, 22seven builds its system around automatic data collection and making the app as easy as possible to use.

These structural differences show three distinct design styles which are behavioural enforcement for YNAB, analytical self-regulation for Wallet, and automation driven access for 22seven. The study shows that no single app currently finds a perfect balance between discipline, ease of use, and automation. This important finding has led to the hybrid design of our proposed application.

By combining structured budget discipline with easy-to-use analysis and selective automation, the new system aims to be both effective and accessible to more people. This comparative review ensures that our proposed framework is built on a real analysis of existing methods instead of just opinions. This strengthens the academic and technical reasons for the chosen design.

Best Features to Include in Our App

The study has found a set of core features to build the foundation of our new app. Using category based budgets with set spending limits is meant to improve money management and stop overspending. Also, the use of live dashboards and long term reports will help users become more aware of their habits by providing clear and useful financial information.

To help users make better choices, the system will send live alerts when spending goes over a limit. This will be paired with an easy way to enter transactions and a place to save digital receipts. These tools will help find a balance between being easy to use and keeping users responsible.

Adding goal tracking modules with clear progress bars will help with long term planning. To make the app more efficient without making users feel disconnected, we will use selective automatic sorting for transactions. Finally, adding simple rewards and game like elements will help users build lasting habits and keep using the app. Together, these features balance discipline, clear data, and smart automation. By using the best parts of the apps we studied and fixing their flaws, our new app creates a strong and inclusive system for modern money management.

Conclusion

The current study shows that digital money management apps play a key role in building financial awareness and better habits. The research into current methods shows that YNAB uses a structured zero based plan that helps with future financial goals. In contrast, Wallet offers deep analysis through detailed dashboard reports and oversight while 22seven focuses on speed through automatic data sorting and links with South African banks.

The comparison highlights the need to balance habit building, clear information design, and smart automation. This research suggests that how well a financial tool works depends on its ability to combine these different design goals.

As a result, our proposed app will include category based budgets, live spending alerts, advanced dashboard reports, and goal tracking tools. By using selective automation, the system aims to create a balanced and inclusive money management tool. This design is built on a real analysis of existing apps. This ensures that the final system fixes the flaws of current platforms while helping users stay involved and financially stable over the long term.

References

Bongomin, G.O.C., Ntayi, J.M., Munene, J.C. and Akol, C.M. (2021) 'Mobile money adoption and financial inclusion in emerging markets', *Journal of African Business*, 22(1), pp. 1–23.

BudgetBakers (2024) *Wallet – Budget & Money Manager*. Available at: https://budgetbakers.com (Accessed: 21 February 2026).

Fernandes, D., Lynch, J.G. and Netemeyer, R.G. (2021) 'Financial literacy, financial education and downstream financial behaviours', *Journal of Consumer Affairs*, 55(1), pp. 80–107.

Kempson, E. and Poppe, C. (2021) 'Can digital tools improve financial capability?', *Journal of Financial Regulation and Compliance*, 29(3), pp. 314–329.

NerdWallet (2026) *The best budget apps for 2026*. Available at: https://www.nerdwallet.com/article/finance/best-budget-apps (Accessed: 22 February 2026).

OECD (2022) *OECD/INFE Policy Guidance on Digitalisation and Financial Literacy*. Paris: OECD Publishing.

PwC (2023) *Global FinTech Report 2023*. Available at: https://www.pwc.com (Accessed: 21 February 2026).

Ranyard, R., McNair, S. and Marriott, P. (2022) 'Digital nudging and financial behaviour change', *Computers in Human Behavior Reports*, 7, 100204.

Sahni, N., Wheeler, S.C. and Chintagunta, P.K. (2023) 'Data-driven financial decision-support systems', *Management Science*, 69(8), pp. 4512–4531.

Vault22 (2024) *22seven (Vault22) platform overview*. Available at: https://www.vault22.io (Accessed: 22 February 2026).

World Bank (2021) *Digital Financial Services and Consumer Behaviour*. Washington, DC: World Bank.

YNAB LLC (2024a) *The four rules*. Available at: https://www.ynab.com/the-four-rules (Accessed: 22 February 2026).

YNAB LLC (2024b) *Features*. Available at: https://www.ynab.com/features (Accessed: 22 February 2026).

Section 2

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Figure 12: Navigation Diagram 77

Section 2 - Planning and Design Purpose Statement

The planning and design section articulates the strategic direction for Zentavo, a hybrid budgeting application developed to support university students and early-career earners in transitioning from reactive spending to intentional financial planning. This section translates the behavioural and usability insights identified in the section 1 research into a structured system design, including the application's design philosophy, core features, and functional and non-functional requirements. By explicitly aligning design decisions with research findings, this section ensures that the development process remains purposeful, evidence-based, and focused on addressing identified gaps in existing budgeting applications.

App Overview

Zentavo is a budgeting application intended to promote responsible financial behaviour among university students and early-career professionals, a population frequently characterised by irregular income patterns and impulse-driven spending habits. Empirical evidence suggests that digital financial management applications can enhance financial literacy and alleviate financial stress by providing accessible tools for monitoring income and expenditure (OECD, 2022; Ruiz-Carhuamaca et al., 2024).

The design of Zentavo was informed by an analysis of three existing budgeting applications: YNAB, Wallet by BudgetBakers, and 22seven. YNAB emphasises behavioural budgeting through proactive allocation rules (YNAB, 2024), Wallet prioritises analytical spending insights via visual expenditure tracking (BudgetBakers, 2024), and 22seven (rebranded as Vault22) focuses on automated financial aggregation and insights (Old Mutual, 2024). Zentavo synthesises these strengths through a hybrid budgeting philosophy that integrates structured allocation and analytical insight, while deliberately limiting automation to maintain user awareness and control

The application introduces several features designed to support behavioural financial management:

- Pre-spending Allocation: Users are able to assign category limits prior to expenditure.
- Dynamic Overspending Indicators: Percentage-based warnings are displayed as category limits are approached, and red visual markers are shown when spending limits are exceeded.
- Financial Insights: Contextualised spending patterns are surfaced for user reflection.

- Interactive Analytics: Visual breakdowns of expenses and spending trends are provided.
- Streak and Level Progression: Users receive achievement badges as rewards for sustained financial tracking.

Existing research indicates that categorisation tools, financial reporting dashboards, and behavioural feedback mechanisms significantly enhance users' ability to manage personal finances (Stefanov et al., 2024). Furthermore, gamification elements, such as milestone-based rewards, have been shown to increase user engagement and promote sustained behavioural change (Bitrián et al., 2021).

Unlike existing applications that focus primarily on either automation, analytics, or strict budgeting structures, Zentavo positions itself as a behaviour-first system that actively guides users towards improved financial habits through continuous feedback and engagement. This design philosophy underpins the functional and non-functional requirements presented in the following sections.

Non-Functional Requirements

Several non-functional requirements are incorporated into the application to ensure system quality and reliability:

- Security: The system, in accordance with secure mobile application development practices, shall encrypt user credentials and safeguard sensitive financial data (Android Developers, 2025).
- Usability: The system shall enable users to create and manage a budget efficiently by providing an intuitive interface (Imawan et al., 2025).
- Performance: The system shall ensure responsive screen loading times and efficient processing on mid-range mobile devices (TestGrid, 2024).
- Reliability: The system shall support offline data storage without the need for internet connectivity, thus enabling continuous expense tracking (Leancode, 2025).
- Scalability: The system architecture shall accommodate optional cloud synchronisation for multi-device access and future feature expansion.

Detailed List of Requirements

The requirements presented in this section are directly derived from the findings of the Part 1 Research Document (Singh et al., 2026) and aligned with the minimum system specifications outlined in the PoE brief. The comparative analysis of YNAB, Wallet, and 22seven identified three key design limitations: insufficient behavioural reinforcement for new users, leading to passive expense tracking rather than active financial management; the absence of integrated gamification features to sustain engagement, despite evidence that structured reward systems support habit formation (Hamari, Koivisto and Sarsa, 2014); and a lack of real-time feedback on overspending, identified as a motivational weakness particularly in Wallet.

Each functional and innovative requirement defined in this section explicitly addresses one or more of these identified gaps, ensuring that the system design is both research-informed and purpose-driven. In addition, non-functional requirements are established in accordance with Android development best practices (Google, 2025a; 2025b), OWASP security standards (OWASP Foundation, 2024), and OECD (2022) policy guidance on digital financial literacy tools, ensuring the system is secure, usable, and scalable.

Functional Requirements

FR-01 User Registration
Requirement • Zentavo shall enable new users to create a personal account by providing a username, password, and password confirmation field.
Validation • Duplicate usernames shall be rejected, and error messages shall not reveal whether a username is already registered, in line with OWASP guidelines (OWASP Foundation, 2024). • Passwords shall be hashed with bcrypt using at least 10 salt rounds. Do not store plaintext passwords in logs or databases. • On first successful registration, display a success message and navigate the user directly to the Dashboard. Zentavo shall automatically populate the account with a default set of spending categories to reduce onboarding friction.

Table 1 FR-01: User Registration

FR-02 User Login
Requirement
Zentavo shall authenticate users using their username and password.
Validation
Lock account access for 15 minutes after five failed login attempts to prevent brute-force attacks (OWASP Foundation, 2024). Error messages shall not reveal whether the username exists in the system. Provide a 'Remember Me' option to keep users signed in until they manually log out. Provide a ‘Forget password’ option to allow the user to change their password. On successful authentication, the user shall be directed to the Dashboard screen.

Table 2 FR-02: User Login

FR-03 Manage Categories
Requirement
Zentavo shall allow users to create, rename, recolour, and delete custom spending categories. Each category shall be assigned a unique name, a colour, and an optional icon.
Behaviour
When a category is deleted, all expenses previously assigned to it are automatically reclassified under the 'Uncategorised' label, ensuring that no historical data is permanently lost. A set of default categories shall be pre-loaded on the first login. Users may edit or delete default categories. Reject duplicate category names within the account shall be rejected with an inline validation message. The category-based structure is directly informed by the design models identified in all three researched applications (Singh et al., 2026).

Table 3 FR-03: Manage Categories

FR-04 Add Expense Entry
Requirement
Zentavo shall allow users to log a new expense by providing at a minimum: an amount (ZAR), a date, a short description, and a category.
Behaviour & Validation
An optional notes field shall be available for further information per entry. Future-dated entries shall be rejected and an inline validation message shall be displayed immediately. Category selection shall be optional and if omitted, the expense shall be classified as 'Uncategorised'. Upon saving, Zentavo shall immediately run an overspending check and update the Dashboard Goal Progress indicator and category bars in real time. This save-and-check feedback loop reflects the alert model observed in YNAB (Singh et al., 2026).

Table 4 FR-04 Add Expense Entry

FR-05 Optional Receipt Photo
Requirement
Zentavo shall permit users to optionally attach a single photograph to any expense entry, either captured via the device camera or selected from the photo gallery.
Behaviour
The photo shall be compressed before storage to conserve the device's storage; the original full-resolution version shall not be retained. Deleting a receipt photo shall not delete the parent expense entry. Where a receipt is attached, a thumbnail icon shall appear inline within the expense list row (see FR-08) and tapping the row shall display the full-size photo without navigating away from the list.

Table 5 FR-05: Optional Receipt Photo

FR-06 Set Monthly Goal
Requirement
Zentavo shall allow users to define a total monthly spending limit referred to throughout the application as the 'Monthly Goal' by specifying an amount and a target calendar month..
Behaviour & Validation
Only one Monthly Goal may be active per calendar month per user account. The Dashboard shall display a real-time Goal Progress indicator employing a traffic-light colour scheme: green for spending below 80%, amber for 80% to 99%, and red at 100% or above. The progress indicator design derives from visual feedback patterns identified in both YNAB and Wallet (Singh et al., 2026).

Table 6 FR-06: Set Monthly Goal

FR-07 Set Category Limits
Requirement
Zentavo shall allow users to define a maximum spending limit per spending category per calendar month, together with an optional alert threshold percentage that triggers an early warning before the limit is fully reached.
Behaviour & Validation
Only one limit per category per calendar month shall be permitted. If a category is already overspent when the limit is saved, the overspending warning shall display immediately without requiring a new expense entry. Category Limits reset automatically at the start of each new calendar month. This requirement is derived from the category-limit design identified in the Part 1 research findings (Singh et al., 2026).

Table 7 FR-07 Set Category Limits

FR-08 View Expense List & Receipt Access
Requirement
Zentavo shall enable users to browse all logged expenses within a user-selected date range, providing search and filter options by category and keyword, as well as multiple sorting options.
Behaviour & Validation
The default view shall display expenses for the current calendar month, sorted in descending order by date (most recent first). When a receipt photo is attached to an entry, a thumbnail icon shall appear inline within the list row. Tapping the row shall open the receipt photo directly within the list view, avoiding navigation to a separate screen. Search results shall update dynamically as the user types, without requiring a separate search submission. Large result sets shall be loaded in a paginated view to maintain scroll performance on lower-end devices.

Table 8 FR-08: View Expense List & Receipt Access

FR-09 View Category Totals
Requirement
Zentavo shall enable users to view the total amount spent per spending category for any user-selected date period, displayed as both a percentage breakdown chart and a ranked list.
Behaviour & Validation
Tapping a category in this view shall navigate to a filtered expense list showing only that category's transactions for the selected period. All percentage values shall sum to 100%; categories with no spending in the selected period shall be hidden. The consolidated category analytics design is informed by the spend-breakdown dashboard architecture identified in 22seven (Singh et al., 2026).

Table 9 FR-09: View Category Totals

FR-10: Local Database Storage
Requirement
Zentavo shall persist all user data, including expenses, categories, monthly goals, category limits, receipt photo references, badges, and streak data, to a local Android Room Database (SQLite), adhering to the MVVM architectural pattern.
Behaviour & Architecture
All database write operations shall execute on a background thread to prevent UI blocking or ANR (Application Not Responding) errors. Zentavo shall operate fully without an internet connection; the local database shall serve as the single source of truth during offline use. Database schema migrations shall be managed using Room's Migration API to preserve all existing user data across app version updates (Google, 2025a).

Table 10 FR-10: Local Database Storage

FR-11 Spending Graph
Requirement
Zentavo shall display an interactive line graph showing daily spending over any user-selected period of up to 365 days, with each spending category represented as a distinct colour-coded data series.
Behaviour & Validation
Horizontal reference lines shall be rendered for the Monthly Goal and any active Category Limits, enabling users to visually assess proximity to each threshold. Days with zero spending shall render as a zero data point rather than a gap, maintaining a continuous line. Users shall be able to pinch to zoom and scroll horizontally across the date axis. Tapping a data point shall show a tooltip with the date, category name, and exact amount (The IIE, 2026a).

Table 11 FR-11: Spending Graph

FR-12 Dashboard
Requirement
Zentavo shall display a real-time progress dashboard as the application home screen, summarising the user's current monthly financial standing at a glance.
Dashboard Content
Total spent this month, remaining Monthly Goal balance, and a colour-coded Goal Progress indicator. Per-category spending bars with Category Limit thresholds marked. A streak counter, a badge summary chip, and the five most recent transactions.
Behaviour
Categories that have exceeded their Category Limit shall be elevated to the top of the category list with a red highlight, ensuring overspending is visible without scrolling. The Dashboard shall update in real time as new expenses are saved and shall also support manual pull-to-refresh. This design integrates best practices identified in Wallet and 22seven during the Part 1 research (Singh et al., 2026).

Table 12 FR-12: Dashboard

FR-13 Gamification elements
Requirement
• Zentavo shall award badges when users reach defined behavioural milestones; each badge shall unlock a functional reward such as an app theme, a motivational insight, or a dashboard upgrade.
One-Time Badges
• First Step: first expense logged. • Receipt Ready: first receipt photo attached. • Planner: all categories allocated for the month.
Monthly Badges
• Budget Hero: monthly goal met. • Category Champion: all category limits met. • Saver's Start: total spending below 50% of the goal.
Streak Badges
• Bronze Streak: 7 consecutive logging days. • Silver Streak: 14 consecutive logging days. • Gold Streak: 30 consecutive logging days.
Rules
• No badge shall be awarded twice within the same qualifying period. • The badge design is based on the empirical gamification framework developed by Hamari, Koivisto, and Sarsa (2014).

Table 13 FR-13: Gamification elements

FR-14 Online Database
Requirement
Zentavo shall allow users to optionally enable Firebase Firestore cloud synchronisation from the Settings screen; cloud sync shall be disabled by default.
Behaviour
When synchronisation is disabled, all core features shall operate fully offline without any degradation in functionality. When sync is enabled, data shall be written to the local Room first, then pushed to Firestore in the background. In the event of a conflict, the most recently modified record shall take precedence (last-write-wins). Unresolvable conflicts shall be flagged in the UI for user review. Failed sync attempts shall retry automatically using exponential backoff (Google, 2025b).

Table 14 FR-14 Online Database

FR-15 Forgot Password
Requirement
Zentavo shall allow users who have forgotten their password to reset it by verifying their registered username and creating a new password, without requiring external communication, consistent with the local-first architecture established in FR-10.
Behaviour
The Forgot Password option shall be accessible directly from the Login screen (FR-02) via a clearly labelled link below the login form. The user shall enter their registered username; if the username does not exist in the local Room Database, an error message shall display without confirming or denying account existence, consistent with the enumeration-prevention principle applied in FR-01 and FR-02 (OWASP Foundation, 2024). Upon valid username entry, the user shall be prompted to enter a new password and to confirm it in a second field; the two fields must match before the submission is accepted. The new password shall be hashed using bcrypt with a minimum of 10 salt rounds before storage in the local Room Database; the plaintext password shall never be stored or logged (OWASP Foundation, 2024). All input fields shall display inline validation messages immediately when a field loses focus, consistent with the usability standard defined in NFR-02. A success message shall display upon completion, and the user shall be automatically navigated to the Login screen. When cloud sync is enabled (FR-14), the updated password hash shall be pushed to Firestore in the background using the write pattern described in FR-14 (Google, 2025b).

Table 15 FR-15: Forgot Password

FR-16 Settings
Requirement
Zentavo shall provide a dedicated Settings screen accessible from the main navigation, consolidating user account management, preference configuration, and application controls in one location. All primary settings shall be reachable within three taps from the Dashboard, in accordance with NFR-02. Settings Items & Behaviour
Edit Profile
Allows the user to update their display username; duplicate usernames shall be rejected with an inline validation message consistent with FR-01.
Change Password
Navigates the user to the password update flow defined in FR-15; the current password must be verified before saving a new one (OWASP Foundation, 2024).
Currency Preference
Enables the user to select a display currency symbol; South African Rand (ZAR) shall be the default, consistent with the financial context established across FR-04, FR-06, and FR-07. All monetary amounts displayed throughout the application shall reflect the selected symbol without altering stored values, supporting the digital financial literacy objectives identified in the OECD (2022) policy guidance.
Budget Notifications Toggle
Enables or disables soft overspending warnings delivered as push notifications. This toggle shall be off by default and produce advisory notifications only, never blocking user actions. This design is consistent with the soft-warning preference identified in group survey findings and the autonomy principle applied in IF-02.
Theme Selection
Displays a list of app themes unlocked through the badge system (FR-13). Users may apply any unlocked theme. Locked themes shall be visible but greyed out, with an indicator of the badge required to unlock them, providing continued engagement incentives consistent with the reward structure described by Hamari, Koivisto, and Sarsa (2014).
Online Database Sync Toggle
Provides the opt-in control for Firebase Firestore cloud synchronisation as defined in FR-14; sync shall be disabled by default, and enabling it shall require a confirmation dialogue. Failed sync attempts shall retry automatically using exponential backoff (Google, 2025b).
Help & Support
Displays a static in-app FAQ and support contact details; this section shall function fully offline, consistent with the offline-first requirement in NFR-03.
About Zentavo
Displays the application version number and a brief description of the app.
Logout
Signs the user out of the current session, clears any active 'Remember Me' session established in FR-02, and navigates to the Login screen. All local Room Database data shall be retained after logout (Google, 2025a).

Table 16 FR-16 Setting

FR-17 Splash
Requirement
Zentavo shall display a branded splash screen on every application launch before navigating the user to their appropriate entry point.
Behaviour
The splash screen shall display the Zentavo logo and application name centred on a background consistent with the active unlocked theme selected in FR-16. The splash screen shall remain visible for a maximum of two seconds, directly supporting the cold-start benchmark defined in NFR-01 and aligned with Android performance best practices (Google, 2025a). Upon expiry, Zentavo shall evaluate the session state: if a valid 'Remember Me' session from FR-02 is active, the user shall be navigated directly to the Dashboard (FR-12); otherwise, the user shall be navigated to the Login screen (FR-02). No user interaction shall be required to dismiss the splash screen; it shall transition automatically. Background initialisation tasks, such as the Room Database warm-up required by NFR-01, shall execute concurrently and silently during this window to ensure no loading delay exceeds the two-second maximum (Google, 2025a). The splash screen constitutes the application entry point referenced in the screen navigation diagram and is consistent with the minimal-friction onboarding standard established in FR-01 (The IIE, 2026a).

Table 17 FR-17 Splash

Innovative Feature Requirements

The innovative feature requirements presented in this section are derived from gaps identified during the Part 1 research phase. These features extend beyond standard budgeting functionality by introducing behaviour-driven mechanisms designed to promote active financial management. IF-01 Pre-Spending Allocation and IF-02 Dynamic Overspending Indicators introduce proactive budgeting and real-time feedback, addressing the lack of behavioural reinforcement and immediate spending awareness identified in existing applications. IF-05 Selective Automatic Sorting addresses the transaction management gap identified in the research (Singh et al., 2026), while maintaining user control over automated processes. Together, these features position Zentavo as a behaviour-first system that integrates feedback, structure, and engagement to support improved financial habits.

IF-01 Pre-Spending Allocation
Requirement
Zentavo shall prompt users to configure Category Limits for the upcoming month before any expenses are logged, operationalising proactive budgeting rather than reactive tracking.
Behaviour
The setup flow shall appear automatically on the first app open of each new calendar month. A 'Copy from Last Month' shortcut shall replicate all previous month allocations in a single tap. The flow is optional and may be skipped; skipping shall not prevent any expense logging. Completing the full allocation awards the 'Planner' badge (FR-13). This feature is derived from YNAB's zero-based allocation model, identified as the most effective behavioural enforcer in the Part 1 research (Singh et al., 2026).

Table 18 IF-01 Pre-Spending Allocation

IF-02 Dynamic Overspending Indicators
Requirement
Zentavo shall display a real-time projected spending warning while the user is entering an expense amount, before the entry is saved.
Behaviour
An amber caution banner shall appear when the projected total would reach the user-defined alert threshold for the selected category. A red warning banner shall appear when the projected total would exceed the Category Limit. Saving shall never be blocked by these warnings; they are informational only and respect user autonomy. After saving, overspent categories shall be elevated to the top of the Dashboard category list with a red highlight. This real-time feedback mechanism reflects the overspending alert functionality identified in Wallet during Part 1 research (Singh et al., 2026).

Table 19 IF-02 Dynamic Overspending Indicators

IF-03 Financial Insights
Requirement
Zentavo shall automatically generate personalised, data-driven spending observations and surface them on the Dashboard and the Analytics screen.
Behaviour
Insights shall only be generated after a minimum of seven days of expense data has been recorded. A maximum of three insights shall be shown on the Dashboard at any time; warning insights shall take priority over positive ones. Insight types include: weekend vs weekday spending patterns, month-on-month category changes, end-of-month projected overspend, and identification of the highest single-day spending (Singh et al., 2026).

Table 20 IF-03 Financial Insights

IF-04 Interactive Analytics Dashboard
Requirement
Zentavo shall provide a dedicated Analytics screen with three selectable views to support deeper financial self-reflection.
Three Views
Spending Trend: a line or stacked area chart of daily spending over a selected period, with pinch-to-zoom support. Category Breakdown: a doughnut chart where tapping a segment filters the expense list to that category. Period Comparison: two user-selected periods side by side with percentage change per category; changes above 20% highlighted in amber.
Export
A summary export option shall allow users to save the Analytics view as a plain-text or CSV file (Singh et al., 2026).

Table 21 IF-04 Interactive Analytics Dashboard

IF-05 Selective Automated Categorisation
Requirement
Zentavo shall support selective automatic sorting for the expense list, providing users control over transaction order while retaining an intelligent default sort behaviour.
Default Behaviour
The default sort order shall be date descending, applied automatically whenever the list is opened or refreshed, consistent with the convention identified across all three researched applications (Singh et al., 2026).
User Options
Date Descending (default) Date Ascending Amount: High to Low Amount: Low to High Category: A–Z
Rules
The selected sort order shall persist for the duration of the session on that screen but shall reset to date descending upon next opening, preventing confusion for casual users. Overspent category entries shall additionally be pinned to the top of the Dashboard transaction summary regardless of sort order, ensuring critical financial information is never buried. This feature addresses the absence of intelligent sort defaults, identified as a usability gap across the researched applications (Singh et al., 2026).

Table 22 IF-05 Selective Automated Categorisation

IF-06 Streak & Level Progression
Requirement
Zentavo shall maintain two independent gamification progression systems: a daily logging streak and a cumulative level system.
Streak System
The streak counter shall increment each calendar day on which at least one expense is logged. It shall be displayed on the Dashboard as a flame icon; the icon turns grey on days with no entry logged yet. The streak resets to 0 if a full calendar day passes without an expense entry.
Level System
Levels never reset. Users earn experience points through badge awards and streak milestones. Progression tiers: Bronze, Silver, Gold, Platinum. An animated achievement card shall display upon level-up. Level progress synchronises with Firestore when cloud sync is enabled (The IIE, 2026a). This dual-system design aligns with Hamari, Koivisto, and Sarsa's (2014) finding that layered reward structures generate more sustained engagement than single-mechanism approaches.

Table 23 IF-06 Streak & Level Progression

Non-Functional Requirements

The non-functional requirements for Zentavo are derived from established best practices within the Android development ecosystem, recognised security frameworks such as OWASP, and current research on usability and digital financial literacy. These requirements define the system's key quality attributes, including performance, usability, reliability, security, scalability, and compatibility, ensuring that the application operates efficiently, securely, and consistently across varying user conditions.

Each requirement is formulated to support the overall system design by enforcing operational standards and enhancing user experience, while remaining aligned with the behavioural and usability insights identified in the research phase. This approach ensures that Zentavo not only meets technical performance benchmarks but also delivers a reliable and user-centred financial management solution in accordance with OECD (2022) policy guidance and Android implementation standards (Google, 2025a; 2025b).

NFR-01 Performance
Requirement
Zentavo shall meet defined response-time and resource benchmarks to ensure a smooth user experience on target devices. Zentavo shall open to the Dashboard screen within two seconds on any device running Android 8.0 (API 26) or above under normal operating conditions. All local Room Database read and write operations shall complete within 500 milliseconds under standard usage loads. The expense list shall implement pagination to maintain scroll performance on datasets exceeding 1000 records.

Table 24 NFR-01 Performance

NFR-02 Usability
Requirement
Zentavo shall adhere to a design standard that minimises cognitive load and ensures accessibility for users without prior experience with budgeting applications. The interface shall follow a minimalist visual design to minimise cognitive load for beginner users, using consistent spacing, typographic hierarchy, and a restrained colour palette. All primary user actions (add expense, view Dashboard, set goal) shall be reachable within three taps from the home screen. All input forms shall display inline validation messages immediately when a field loses focus, rather than only on form submission.

Table 25 NFR-02 Usability

NFR-03 Reliability
Requirement
Zentavo shall remain stable and data-safe under both normal and adverse operating conditions, including complete offline use. Zentavo shall remain fully operational without an internet connection; all core features shall function exclusively using local Room Database storage, with no functionality disabled during offline use. Zentavo shall process unexpected database errors gracefully, displaying a user-facing error message rather than crashing or producing an ANR dialogue. Data entered by the user shall not be lost due to an application crash or device interruption occurring during a save operation.

Table 26 NFR-03 Reliability

NFR-04 Security
Requirement
Zentavo shall protect all user credentials and financial data in both local storage and cloud environments. All user passwords shall be hashed using bcrypt with a minimum of 10 salt rounds prior to storage; plaintext passwords shall never be persisted (OWASP Foundation, 2024). Zentavo shall ensure that all user data stored in Firebase Firestore is isolated per user account via server-side Firestore Security Rules, preventing cross-user data access regardless of client-side behaviour. Zentavo shall not log or expose sensitive user data, including passwords, email addresses, or financial amounts, in system logs or crash reports.

Table 27 NFR-04 Security

NFR-05 Scalability
Requirement
Zentavo's architecture shall accommodate future growth in users, features, and data volume without requiring significant structural rework. The system architecture shall support future modular feature expansion, including additional badge types, new analytics views, and gamification mechanics, without restructuring the core Room or Firestore database schemas. The Firestore data model shall use per-user subcollections to ensure that scaling the total user base does not degrade query performance or data isolation for any individual user account.

Table 28 NFR-05 Scalability

NFR-06 Compatibility
Requirement
Zentavo shall run correctly across the range of Android devices and screen configurations encountered by its target users. Zentavo shall be compatible with Android devices running API level 26 (Android 8.0 Oreo) and above. The application layout shall support both portrait and landscape orientations without layout degradation or data loss. Zentavo shall run correctly on both Android emulators during development and on physical Android devices (The IIE, 2026a).

Table 29 NFR-06 Compatibility

UI and Gantt Chart

Screen Descriptions

Onboarding:

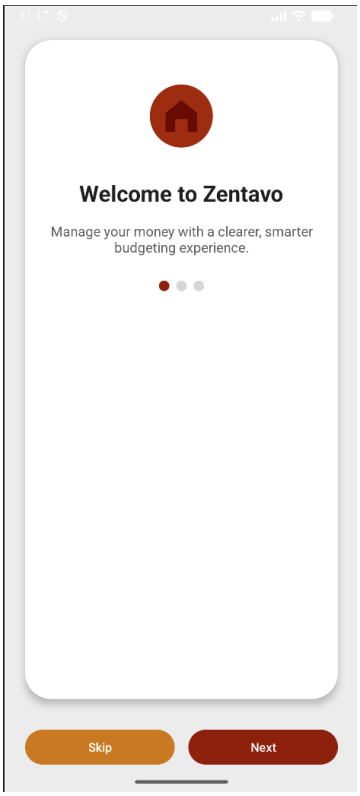

Figure 16: Onboarding Screen

The **onboarding screen** represents the user's first interaction with the application and is designed to introduce the core purpose and value of the budgeting system. Its primary function is to familiarise users with key features, including financial tracking, pre-spending planning, and motivational mechanisms such as rewards. This screen supports the system's behavioural design objectives by reinforcing features such as category allocation (IF-01) and gamification elements (FR-13), as identified in the requirements specification. By presenting concise information through a structured, multi-step flow, the onboarding process minimises cognitive load and supports ease of understanding for first-time users. This ensures a smooth transition into the application while preparing users for effective engagement with the system's core functionalities.

Login and Registration:

The **login screen** provides secure entry into the application and functions as the primary authentication gateway. Users are required to input their credentials to access their financial dashboard, with additional navigation options to registration and password recovery features. This screen directly implements the system's authentication requirements, including user login (FR-02), registration (FR-01), and password recovery (FR-15), ensuring controlled and secure access to user data. The design prioritises simplicity and clarity to reduce user friction during onboarding, supporting usability principles while maintaining robust access control. This ensures a seamless user experience while safeguarding sensitive financial information.

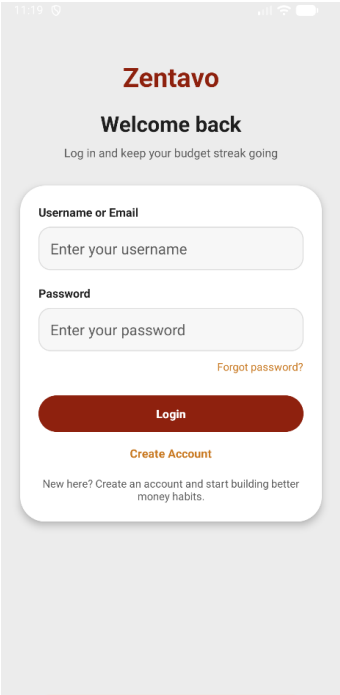

Figure 17: Login Screen

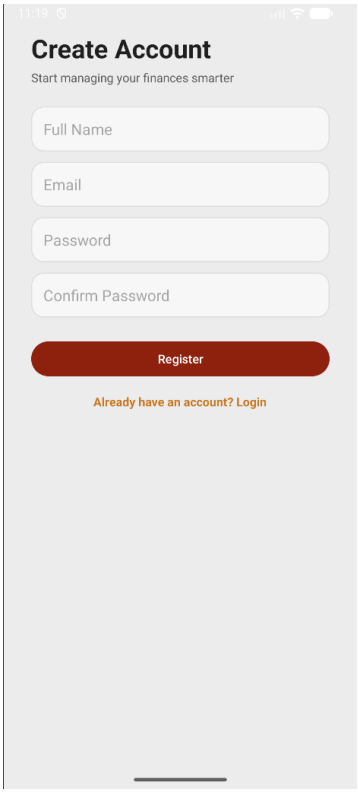

Figure 18: Registration Screen

The **registration screen** enables new users to create an account by providing essential authentication information, including a username and password. This screen directly implements the user registration requirement (FR-01) and forms a critical component of the system's onboarding process. Its inclusion ensures functional requirement traceability while supporting secure account creation and initial system access.

Although currently implemented as a prototype, the screen accurately represents the intended registration workflow and interaction design. The interface is deliberately kept simple and structured to minimise cognitive load and reduce user friction during account creation. Additionally, clear navigation back to the login interface is provided to support usability and prevent user disorientation. This design ensures a smooth and accessible onboarding experience aligned with the system's usability objectives.

Forgot Password:

The **forgot password** screen supports account recovery by enabling users to reset their credentials through a secure password update process. This screen directly implements the password recovery requirement (FR-15) and forms an integral part of the system's authentication framework. By providing a fallback mechanism for credential-related issues, it enhances system reliability and reinforces user trust in the application's security.

Although currently implemented as a simulated feature, the screen accurately represents the intended recovery workflow and interaction design. The interface is structured to guide users through the reset process with clear input validation and feedback, reducing user confusion and preventing errors. This design supports both usability and security objectives by ensuring that account recovery is accessible, controlled, and aligned with secure authentication practices.

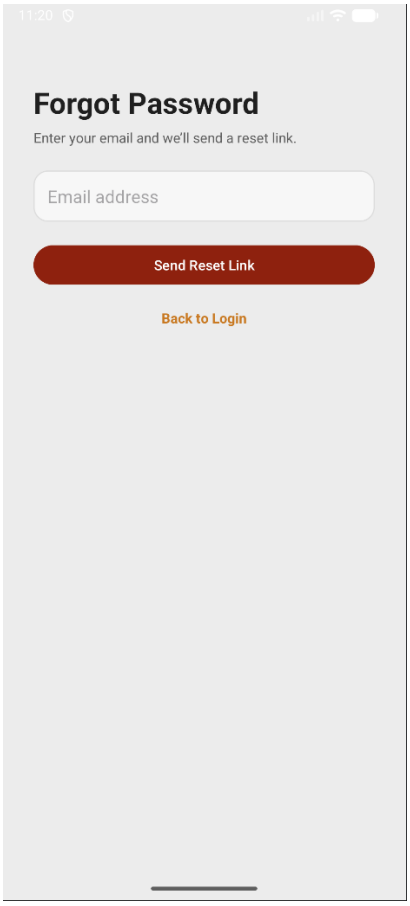

Figure 19: Forgot Password Screen

Dashboard:

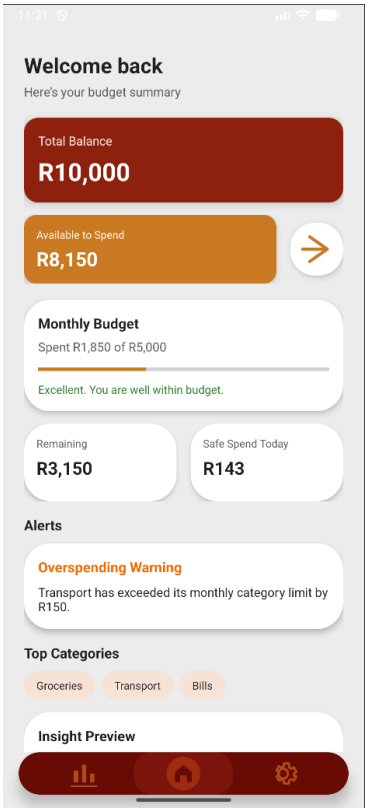

Figure 20: Dashboard Functions

The **dashboard** functions as the central overview of the application, serving as the primary interface through which users monitor their financial status. Its purpose is to provide a high-level summary of key information, including total monthly spending, remaining budget, recent transactions, and motivational insights. This screen directly implements the dashboard requirement (FR-12), integrating data from multiple system components such as expense tracking (FR-04), category limits (FR-07), and monthly goals (FR-06).

The dashboard is designed to update in real time as new expenses are recorded, ensuring that users receive immediate feedback on their financial behaviour. It incorporates behavioural finance principles through the use of progress indicators, visual cues, and feedback messages that encourage responsible spending and reinforce budgeting habits. By prioritising critical information and elevating overspending indicators, the design supports rapid decision-making while maintaining clarity and usability.

Finance:

The **finance screen** provides detailed analytical insights into user spending and serves as the primary interface for financial data exploration. It includes visualisations such as pie charts and trend graphs, directly implementing the spending analytics and category breakdown requirements (FR-09, FR-11), enabling users to interpret their expenditure patterns effectively.

In addition to data visualisation, the screen functions as a navigation hub for key financial actions, including adding new expenses (FR-04), accessing the monthly planner (FR-06, FR-07), and reviewing historical transactions (FR-08). This integration of analytics and functionality allows users to both analyse and act on their financial data within a single interface.

The design supports informed decision-making by presenting complex financial information in a clear and visually structured manner, reducing cognitive load while enhancing user awareness. By combining analytical insight with direct action pathways, the finance screen reinforces the system's behaviour-driven approach to financial management.

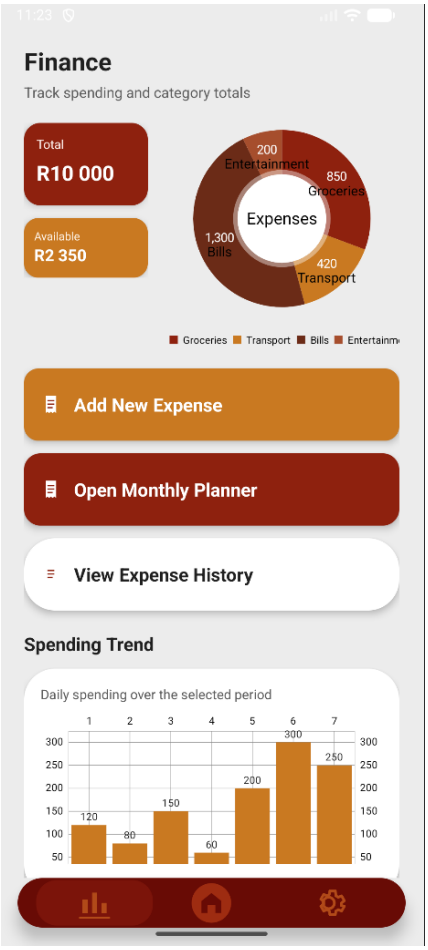

Figure 21: Finance Screen

Add Expense:

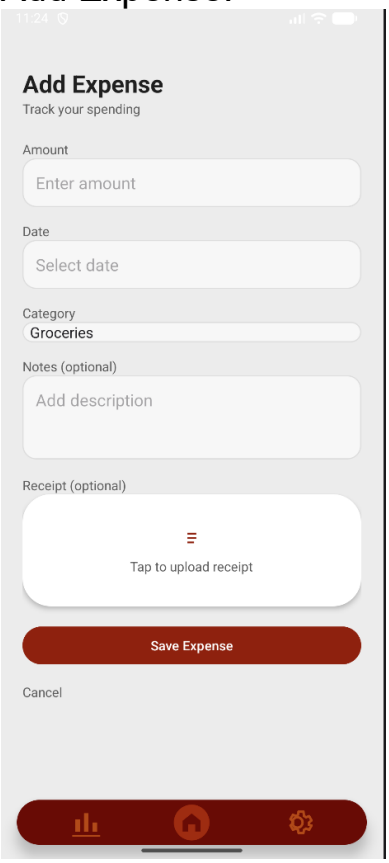

Figure 22: Add Expense Screen

The **add expense** screen enables users to record financial transactions by entering key details such as amount, date, category, and optional notes or receipt images. This screen directly implements the expense entry requirement (FR-04) and integrates with category management and receipt handling features (FR-03, FR-05), forming a core component of the system's data input functionality.

A key feature of this screen is the implementation of dynamic overspending warnings (IF-02), which provide real-time, predictive feedback on how a transaction will impact category limits (FR-07). This allows users to assess the consequences of their spending before confirming the transaction, supporting proactive budgeting behaviour.

The interface is designed to guide users through a structured input process while minimising cognitive load through clear fields and immediate validation feedback. By combining data entry with real-time behavioural feedback, the screen reinforces the system's behaviour-driven approach to financial management and encourages more informed spending decisions.

Monthly Planner:

The **monthly planner** screen facilitates pre-spending financial planning by allowing users to define a monthly budget and allocate category limits. This screen directly implements the budgeting and category allocation requirements (FR-06, FR-07), forming the foundation of the system's structured financial planning functionality.

It operationalises the pre-spending allocation concept (IF-01), which requires users to assign financial limits before expenditure occurs. This approach encourages proactive budgeting behaviour and reduces the likelihood of reactive overspending by establishing clear financial boundaries in advance.

The interface presents category limits and budget allocations in a structured and visually accessible format, enabling users to plan their finances with clarity while minimising cognitive load. By shifting financial decision-making to a pre-spending context, the monthly planner reinforces disciplined financial behaviour and supports the system's behaviour-driven design philosophy

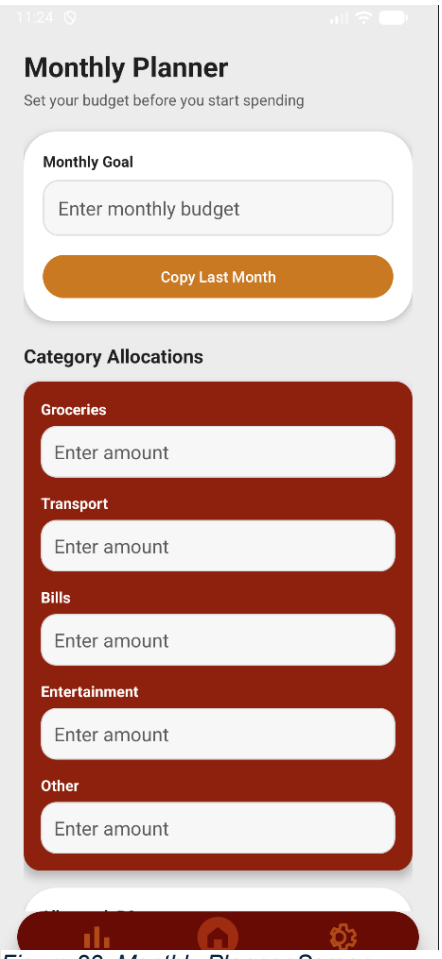

Figure 23: Monthly Planner Screen

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Expense History:

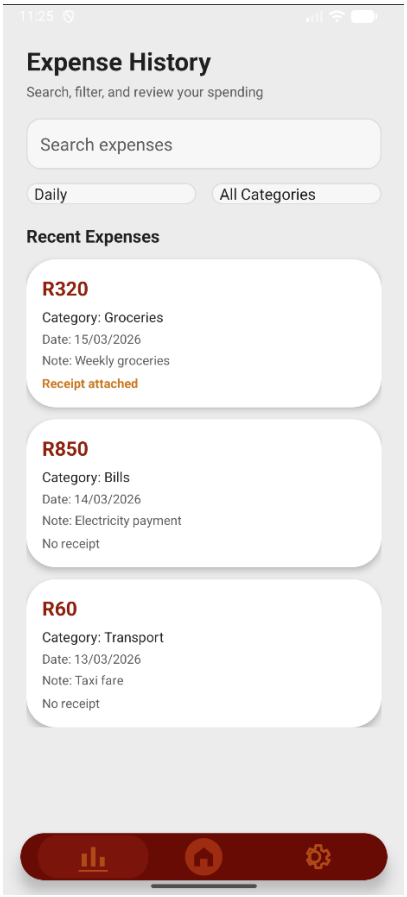

Figure 24:Expense History Screen

The **expense history** screen allows users to review previously recorded transactions through searchable and filterable lists. This screen directly implements the transaction history requirement (FR-08) and integrates with category classification and expense tracking features (FR-03, FR-04), ensuring that all financial data remains accessible and organised.

By presenting detailed expense information, including category, date, and notes, the screen enhances transparency and enables users to identify spending patterns over time. The inclusion of search and filtering functionality supports efficient data retrieval, allowing users to focus on specific categories or time periods for deeper analysis.

This screen plays a critical role in retrospective financial evaluation by enabling users to reflect on past behaviour and identify trends that may require adjustment. By supporting structured financial review, it reinforces the system's behavioural objective of promoting awareness and continuous improvement in spending habits.

Gamification:

The **gamification** screen introduces motivational elements such as streak tracking, achievement badges, and level progression, directly supporting the implementation of gamification elements (FR-13) and streak and level progression mechanisms (IF-06). This screen is designed to reinforce consistent financial tracking behaviour by providing users with clear visual indicators of progress and achievement.

By rewarding actions such as regular expense logging, adherence to category limits, and sustained budgeting activity, the system establishes a positive reinforcement loop that encourages continued engagement. These mechanisms promote habit formation by linking disciplined financial behaviour with measurable rewards and progression milestones.

The interface presents user achievements in a structured and visually engaging format, enabling users to track their performance over time. By integrating behavioural reinforcement into the core budgeting functionality, the gamification screen enhances long-term user retention and contributes to the system's objective of fostering sustained financial responsibility

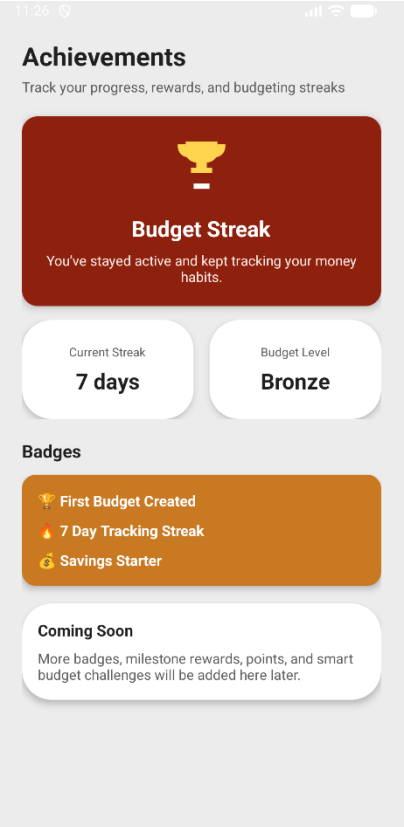

Figure 25: Gamification Screen

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Settings:

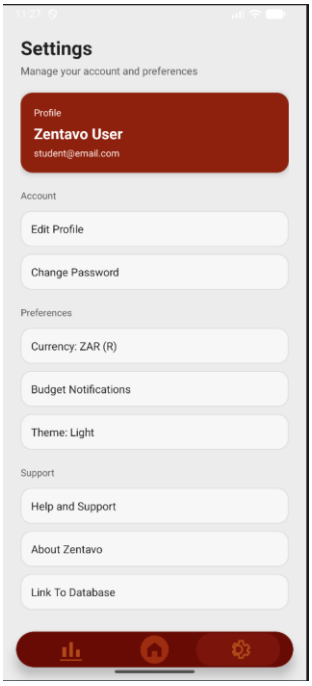

Figure 26: Settings Screen

The settings screen functions as a configuration hub where users can manage application preferences and system features, directly supporting settings management functionality (FR-16). This screen also aligns with supporting functional requirements, including database interaction (FR-14), password management and recovery (FR-15), and authentication-related behaviour (FR-02), where relevant to user account control.

In addition, the screen supports key non-functional requirements, particularly usability (NFR-02) and security (NFR-04), by enabling users to customise their experience and manage sensitive account settings. Features such as notification preferences, password updates, and general system configuration enhance accessibility and ensure secure interaction with the application.

By providing a centralised and structured interface for system configuration, the settings screen improves user control and adaptability over time. This contributes to sustained user engagement while reinforcing the system's user-centric design and compliance with established mobile application best practices

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Navigation Diagram

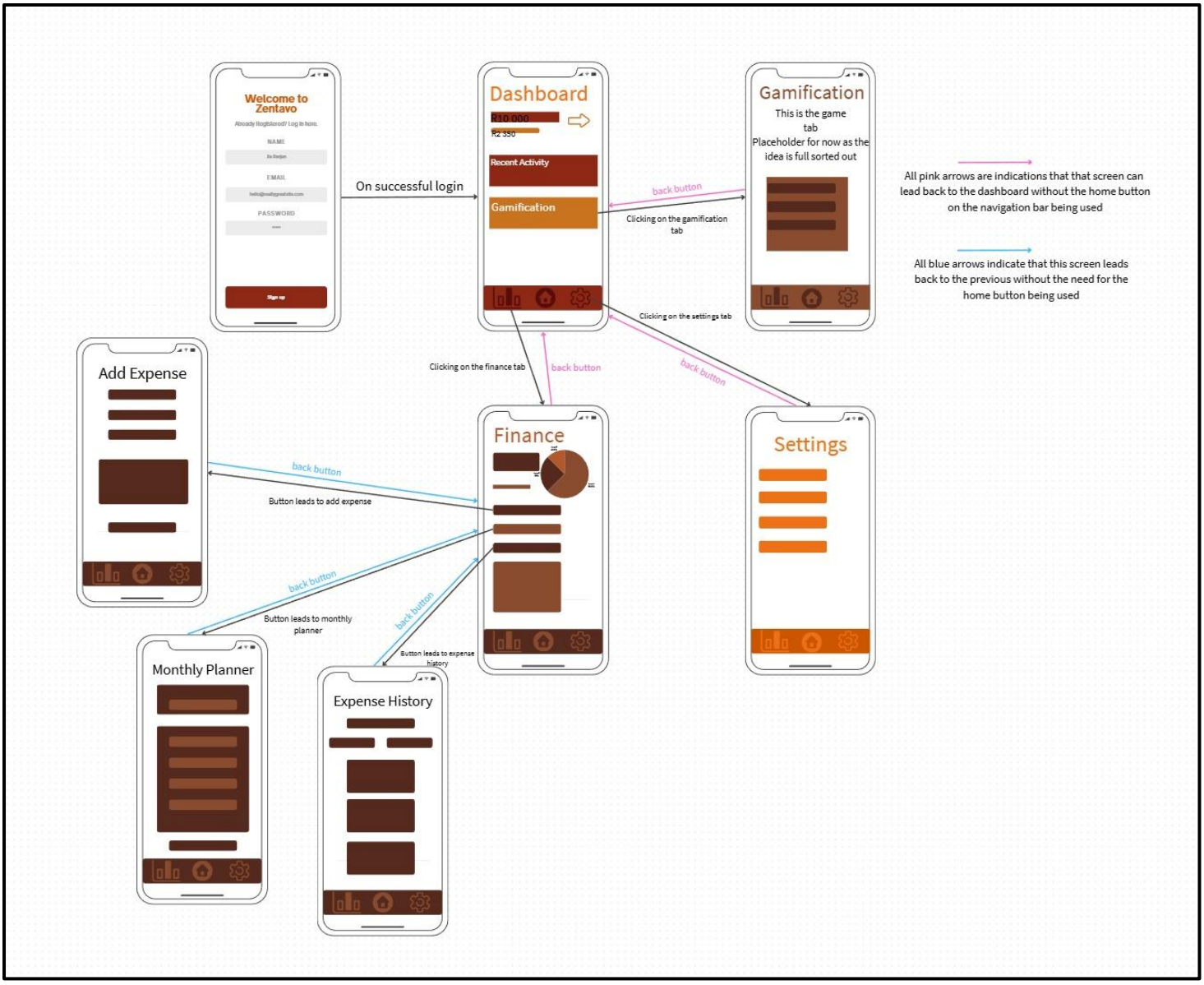

Figure 27: Navigation Diagram

The app will begin at the login page upon startup. After the user logs in they will be sent to the home page/dashboard where they will get a comprehensive overview of their finances, recent activity and any achievements they may have earned. The bottom navigation menu will allow the user to go between the dedicated finance page, home menu and settings. This is to allow ease of use and to separate any clutter that may have been on the home page otherwise. The game tab has a dedicated button on the home page to take the user there when requested.

Gnatt Chart

The project plan is structured across seven phases, following a logical progression from research and requirement definition to system development, testing, and final submission. Each phase is aligned with key deliverables identified in the project scope, ensuring that design, implementation, and evaluation activities are completed in a structured and iterative manner. The timeline reflects a balanced distribution of workload, allowing for development, refinement, and testing to ensure system quality and reliability.

Phase 1: Research & Requirements

Task	Week 1	Week 2	Week 3
User needs analysis	✓		
Research existing budgeting apps	✓	✓	
Identify system requirements		✓	✓
Define core app features		✓	✓
Technology selection			✓
Team role allocation	✓		

Phase 2: Planning & System Design

Task	Week 4	Week 5	Week 6
Navigation diagram design	✓		
UI wireframe design	✓	✓	
Database structure planning		✓	✓
Budget logic planning		✓	✓
Gamification concept planning			✓
Project timeline planning	✓	✓	✓

Phase 3: UI Development

Task	Week 7	Week 8	Week 9
Login screen UI	✓		
Dashboard UI	✓	✓	
Finance screen UI		✓	✓
Settings screen UI		✓	✓
Gamification UI prototype			✓
Bottom navigation implementation	✓	✓	✓

Phase 4: Core Feature Development

Task	Week 10	Week 11	Week 12	Week 13
Expense entry functionality	✓	✓		
Category management		✓	✓	
Budget calculation logic		✓	✓	✓
Local data storage integration			✓	✓
Receipt upload functionality (prototype)			✓	✓
Navigation logic refinement	✓	✓	✓	✓
User authentication implementation (login, registration, recovery)		✓	✓	✓

Phase 5: Integration & Testing

Task	Week 14	Week 15
Feature integration	✓	✓
Usability testing	✓	✓
Bug identification and resolution	✓	✓
UI responsiveness and device compatibility testing	✓	✓
Data persistence testing	✓	✓

Phase 6: Enhancement & Optimisation

Task	Week 16	Week 17
UI visual refinement	✓	✓
Performance optimisation	✓	✓
Gamification feature refinement and implementation	✓	✓
User feedback improvements	✓	✓
Performance benchmarking and optimisation	✓	✓

Phase 7: Documentation & Submission

Task	Week 18	Week 19	Week 20
Compile research document	✓		
Finalise design documentation	✓	✓	
Prepare demo		✓	✓
Final system testing		✓	✓
Final submission			✓

Readiness Confirmation

All mandatory POE application features have been identified and aligned with the system terminology used throughout the design process, including constructs such as Category Limit, Monthly Goal, and Dashboard (IIE, 2026b). This ensures consistency between the requirements, user interface design, and system architecture. The planning phase therefore establishes a structured and coherent foundation for the subsequent development stage, during which the defined functional requirements and interface specifications will be implemented and validated.

Conclusion

The design strategy for Zentavo is grounded in contemporary research and informed by a comparative analysis of existing financial budgeting applications. By integrating the behavioural allocation approach of YNAB, the analytical insight capabilities of Wallet, and the automation principles of 22seven, the proposed system adopts a hybrid design philosophy that combines structure, feedback, and user engagement.

This integrated approach directly addresses the limitations identified in the research phase, particularly the lack of behavioural reinforcement, limited user engagement, and insufficient real-time feedback in existing solutions. As a result, Zentavo is positioned to support users in developing sustainable financial habits while enhancing long-term financial decision-making (YNAB, 2024; BudgetBakers, 2024; Old Mutual, 2024).

References

Android Developers. 2025. *Security best practices for Android apps*. Available at: https://source.android.com/docs/security/best-practices [Accessed 6 March 2026].

Bitrián, J., Buil, C. and Catalán, J. 2021. 'Making finance fun: the gamification of personal financial management apps'. *International Journal of Bank Marketing*. Available at: https://selfdeterminationtheory.org/wp-content/uploads/2024/03/2021_BitrianBuilCatalan_IJBM.pdf [Accessed 5 March 2026].

BudgetBakers. 2024. *Wallet – Budget & Money Manager*. Available at: https://budgetbakers.com/en/products/wallet/ [Accessed 6 March 2026].

Google. 2025a. *Save data in a local database using Room*. Available at: https://developer.android.com/training/data-storage/room [Accessed 14 March 2026].

Google. 2025b. *Cloud Firestore documentation*. Available at: https://firebase.google.com/docs/firestore [Accessed 14 March 2026].

Hamari, J., Koivisto, J. and Sarsa, H. 2014. 'Does gamification work? A literature review of empirical studies on gamification'. In: *Proceedings of the 47th Hawaii International Conference on System Sciences*. IEEE, pp. 3025–3034.

Imawan, R., Putra, W., Alqahtani, R., Milakis, E. and Dumchykov, M. 2025. 'Enhancing financial literacy in young adults: An Android-based personal finance management tool'. *Journal of Hypermedia & Technology-Enhanced Learning*.

Available at: https://edutech-journals.org/index.php/j-hytel/article/view/166/87 [Accessed 7 March 2026].

Leancode. 2025. *Offline mobile app design: principles and architecture*. Available at: https://leancode.co/blog/offline-mobile-app-design [Accessed 6 March 2026].

OECD. 2022. *OECD/INFE guidance on digital delivery of financial education*. Paris: OECD Publishing.

Old Mutual. 2024. *22seven rebrands to Vault22*. Available at: https://www.oldmutual.co.za/news/22seven-rebrands-to-vault22/ [Accessed 5 March 2026].

OWASP Foundation. 2024. *Password storage cheat sheet*. Available at: https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html [Accessed 14 March 2026].

Ruiz-Carhuamaca, Y., Yauricasa-Seguil, M. and Morales-Arevalo, L. 2024. 'Design of a mobile learning app for financial literacy'. *International Journal of Advanced Computer Science and Applications*, 15(12). Available at: https://thesai.org/Downloads/Volume15No12/Paper_10-Design_of_a_Mobile_Learning_App_for_Financial_Literacy.pdf [Accessed 4 March 2026].

Singh, N., Kubheka, M., Marwane, O., Marota, L.R. and Tema, M.L. 2026. *PROG7313 Portfolio of Evidence — Part 1: Research document*. Johannesburg: The Independent Institute of Education.

Stefanov, R., Ivanova, D. and Petrov, P. 2024. ‘The impact of personal finance applications on financial decision-making’. *TEM Journal*, 13(3), pp. 2066–2075. Available at: https://www.temjournal.com/content/133/TEMJournalAugust2024_2066_2075.pdf [Accessed 6 March 2026].

TestGrid. 2024. *Mobile app performance testing: best practices*. Available at: https://testgrid.io/blog/mobile-app-performance-testing/ [Accessed 7 March 2026].

The Independent Institute of Education. 2026a. *PROG7313 Programming 3C / Open Source Coding (Introduction): Module manual*. Johannesburg: IIE.

The Independent Institute of Education. 2026b. *PROG7313 / OPSC7311 Portfolio of Evidence (POE)*. Johannesburg: IIE.

YNAB. 2024. *The YNAB Method*. Available at: https://www.ynab.com/ynab-method [Accessed 5 March 2026].